NaviCav: A WEB–MOBILE ROUTE PLANNER FOR MULTI-MODAL COMMUTING IN CAVITE USING MONTE CARLO–GUIDED GENETIC OPTIMIZATION.

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MCGA	Monte Carlo–Guided Genetic Algorithm
GA	Genetic Algorithm
MC	Monte Carlo Simulation
RAPTOR	Round-Based Public Transit Optimized Router
LPTRP	Local Public Transport Route Plan
PUJ	Public Utility Jeepney
PUVMP	Public Utility Vehicle Modernization Program
GTFS	General Transit Feed Specification
P2P	Point-to-Point
SUS	System Usability Scale
TAM	Technology Acceptance Model

CHAPTER I

INTRODUCTION

Background and Rationale of the Study

The public transportation system in the Philippines continues to face significant pressures, despite recent reforms and public expenditure on sustainable infrastructure. Informal and fragmented transport services such as jeepneys, tricycles, and multicabs persist in many travel corridors, often without defined service intervals or pre-established stops. These service characteristics lead to long and unpredictable travel times, unpredictable headways, and localized congestion, all of which reduce commuter utility while increasing emissions and safety risks (UP National Center for Transportation Studies [UP NCTS], 2025; Gaspay et al., 2023).

The province of Cavite illustrates the mismatch between accelerating travel demand and existing transport capacity. Rapid population and economic growth in key urban centers such as General Trias and Trece Martires has increased travel demand along major corridors, including Governor's Drive and Aguinaldo Highway. The result has been:

- Severe congestion at critical intersections
- Uncoordinated terminals that hinder smooth transfers
- Lengthened total trip times and reduced reliability of daily commutes

Local planning documents point to these challenges in their findings. The Trece Martires Local Public Transport Route Plan (LPTRP) flagging the problem was relatively increasing traffic congestion along major roads and intersections. The report also notes the ongoing use of old-model public utility vehicles (PUVs), which cause air and noise pollution, as one of their suggestions was to replace them by electric or newer model transport (City Government of Trece Martires, 2024).

While local government efforts and infrastructure initiatives have resolved some issues, commuters still do not have complete and real-time routing. The routing has not taken into account the variability of the informal modes (DPWH, 2023; Provincial Government of Cavite, 2023; Lara & Perido, 2024).

The City Government, specifically General Trias and Trece Martires municipalities, are the chief institutional partners to this work. As local governments acting as executive branches, they are charged with a range of mobility dilemmas including resolving congestion, and improving the residents' mobility experience, through town planning. As part of the planning capacity, the City Planning and Development Office designs plans of transport and infrastructure within its Local Public Transport Route Plan (LPTRP), the Traffic Management Office implements and enforces such plans, including monitoring the range of issues related to colorum transports, and the Information and Communications Technology Department (ICTD) manages custodianship of digital systems. Clearly, as they fulfill this mandate, city governments rank well as an important client for a commuter-focused route planner that streamlines information

and evidence-based decision making, regulatory enforcement, and sustainable urban mobility.

Current transport planning practices in Cavite are constrained by fragmented datasets, predominantly manual procedures, and timetable-based data formats that are poorly aligned with informal operations. International and national transit data standards and common journey-planning platforms typically assume formalized routes and stable schedules, which reduces their applicability in contexts dominated by informal, stop-on-demand services (Google Transit Partners, n.d.; Transformative Urban Mobility Initiative data examples). Consequently, commuters and planners lack accessible tools that combine real-time data, reliability measures, and multi-criteria decision support that are tailored to local modal behavior and commuter expectations (Strenitzerová & Stalmachová, 2021; Fraginal et al., 2024).

Contemporary research supports the use of hybrid computational frameworks to address the dual challenges of uncertainty and multiple optimization objectives in multimodal route planning. Monte Carlo simulation offers a principled approach for representing variability in travel times, dwell times, and demand, while Genetic Algorithms provide a population-based search mechanism capable of optimizing several conflicting criteria simultaneously, including travel time, waiting time, number of transfers, operating cost, and emissions (Peng, Mo, & Liu, 2021; Liu et al., 2025;

Wang et al., 2024). Hybrid Monte Carlo and Genetic Algorithm frameworks have been shown to improve solution robustness under stochastic inputs and to produce resilient route options for multimodal contexts in a range of international studies (Chen et al., 2024; Guo et al., 2024).

Environmental considerations further strengthen the rationale for localized routing systems that incorporate carbon metrics. Research demonstrates that carbon-aware routing can reduce fuel consumption and CO_2 emissions when paired with optimization techniques (Zrigui et al., 2023; Vergel et al., 2024). Incorporating such metrics for Cavite's common modes such as jeepneys, tricycles, and multicabs that would enable commuters and planners to consider sustainability in travel choices and policy evaluation.

Additionally, software quality requirements play an important role in commuter-facing systems. International quality models refer product characteristics of functionality, usability, reliability, and efficiency (ISO/IEC 25010; Maia et al., 2021; Putra et al., 2024). For a localized web-and-mobile planner to be practically adopted, it must deliver functionally correct, reliable, and user-friendly performance within acceptable time constraints.

Specifically, this study addresses four interrelated problems:

1. High variability and unpredictability in travel times, leading to long and unreliable commutes.

2. Fragmented modal networks and inadequately coordinated transfer facilities, which extend trip durations.
3. The absence of commuter-facing tools that integrate real-time data, uncertainty modelling, and multi-criteria optimization suitable for informal modes.

4. The lack of eco-aware routing metrics that would enable commuters and planners to account for emissions in route choice and planning (Lara & Perido, 2024; Fraginal et al., 2024; Vergel et al., 2024).

In response to these challenges, the present project proposes the design, development, and evaluation of a Monte Carlo–Guided Genetic Optimization (MCGA) web-and-mobile route planner for Cavite. The system aims to integrate formal and informal transport modes, model uncertainty in travel times and transfers, optimize multiple performance criteria simultaneously, and comply with established software quality standards so it can provide reliable, usable, and practically useful decision support for both commuters and provincial planners.

This study is aligned with the United Nations Sustainable Development Goals (SDGs). By promoting sustainable transport planning and reduces dependence on unproductive and informal travel patterns to support SDG 11 (Sustainable Cities and Communities), promotes SDG 9 (Industry, Innovation, and Infrastructure) with the application of advanced optimization and ICT-based systems for transport planning, and promotes SDG 13 (Climate Action) through the integration of eco-aware routing to address emissions from transport.

Objectives of the Study

The general objective of this is to develop a web–mobile multi-modal route planning system that applies Monte Carlo–Guided Genetic Optimization (MCGA) for adaptive and efficient generation of near-optimal commuting routes under multiple constraints. Moreover, the study would like accomplish the following specific objectives:

1. To design and develop a web–mobile route planning system architecture that integrates multi-modal commuting data sources in Cavite addressing the current fragmentation of public transportation.
2. To implement Monte Carlo–guided genetic optimization as the core algorithm for generating adaptive and near-optimal commuting routes.
3. To apply a Search-Based Software Engineering (SBSE) approach by formulating route planning as a search problem and optimizing solutions under constraints such as travel time, cost, and carbon footprint. Thus, filling the gap in commuter-facing tools that incorporate eco-aware and multi-criteria decision-making
4. To evaluate the system's computational performance in terms of algorithm efficiency, scalability, and adaptability to dynamic commuting conditions.

5. To assess software quality based on ISO/IEC 25010 standards, focusing on functionality, usability, reliability, and efficiency through expert review and end-user validation.

Significance of the Study

Efficient mobility is one of the central challenges facing Cavite today, as commuters and city leaders contend with fragmented transport networks, congestion, and rising travel demand. Addressing these challenges requires localized, data-driven solutions that integrate both formal and informal modes of transport. The present study responds to this need by developing a route planning system that applies advanced computational methods to improve daily travel and inform long-term mobility planning.

Commuters. The primary beneficiaries are Cavite's commuters, who will gain accurate route suggestions, fare breakdowns, terminal information, and travel alternatives. These features aim to reduce travel time, improve accessibility, and lower commuting costs, leading to more reliable and convenient daily travel.

City Government (General Trias, and Trece Martires). As the main client, the mayor and local government units will gain a collaborative decision-support tool for improving mobility and addressing congestion. Within this structure:

- The **City Planning and Development Office** will use the system to design and evaluate route plans.

- The **Traffic Management Office (TMO)** will strengthen its regulatory function by directing commuters toward legal terminals and hubs, thereby discouraging reliance on colorum operators.

Academic and Research Community. For the proponents, the project provides an opportunity to apply advanced computational methods such as Monte Carlo simulation, Genetic Algorithms, and Search-Based Software Engineering (SBSE) in a real-world context. Future researchers may build on this work to refine or expand optimization models for intelligent transportation systems across different contexts.

Local Communities and the Environment. By incorporating carbon metrics into route planning, the system encourages eco-aware travel decisions and supports the reduction of CO_2 emissions. These benefits contribute to Cavite's sustainability objectives while aligning with national and global commitments to climate-resilient and sustainable transport.

Scope and Limitation

This study aims to develop NaviCav: A Web–Mobile Route Planner for Multi-Modal Commuting in Cavite Using Monte Carlo–Guided Genetic Optimization. The system focuses on optimizing routes for public transport modes commonly used in Cavite which are jeepneys, multicabs, and tricycles by providing multi-criteria route options (time, cost, transfers, and carbon footprint). The initial geographic coverage will be limited to the adjacent cities of Trece Martires and General Trias which represent high-demand commuter corridors (General Trias City Government, 2020;

Highways Today, 2024; Lara & Perido, 2024; Philippine Statistics Authority CALABARZON, 2023; Provincial Government of Cavite, 2023; Zafra, Briones, & Alampay, 2023). This narrower scope enables more focused data collection, manageable implementation, and clearer evaluation, with the potential for expansion to other areas of Cavite in future studies.

The project is intended for commuters who rely on public transport for daily mobility. It does not cover route optimization for private cars, point-to-point shuttles, or ride-hailing services. Moreover, the system will not provide live vehicle tracking but will rely on structured data (routes, stops, fares, estimated travel times) and simulation models to generate results.

Limitations of the study include: (1) reliance on available datasets from local government offices and field surveys, which may be incomplete or inconsistent; (2) exclusion of inter-provincial routes beyond Cavite; (3) prototype-level deployment, meaning the system will be tested for functionality, usability, and algorithmic performance but not rolled out as a full-scale commercial platform.

CHAPTER II

REVIEW OF RELATED LITERATURE

The State of Public Transportation in the Philippines

Public transportation in the Philippines remains heavily strained despite ongoing reforms. In general, the country "lags many neighboring countries in infrastructure development and is notorious for challenging traffic conditions and long commutes" (U. S. International Trade Administration [USITA], 2024). Metro Manila, which inhabits a population of 13.5 million, and can be higher during daytime, accounts for 37% GDP while only occupying <1% of the Philippines' land area, has the highest congestion level among 278 cities with a population greater than 5 million and is also subject to low public transport quality and decreasing ridership (Gaspay et al., 2023). About 240, 000 jeepney units that are registered nationwide contribute to 40% of land-based transport trips despite 75% of the units above 15 years old (UP National Center for Transportation Studies [UP NCTS], 2025). According to UP National Center for Transportation Studies (2025), this poses serious safety risks and emission problems as

PUJs account for 34% of the overall national greenhouse gas emissions. In short, decades of informal, fragmented transport have left the system overcrowded, unreliable, and outdated.

Cavite, one of the fastest growing (Philippine Statistics Authority CALABARZON [PSA CALABARZON], 2023) and most populated provinces (Philippine Statistics Authority CALABARZON [PSA CALABARZON], 2021) in

the country, illustrates the challenges faced by areas located on the outskirts of Metro Manila. Arcadis (2023) noted that, "Cavite and Laguna have been at the forefront of industrial development ", but their transportation has "not kept pace with rapid growth", which results in traffic congestions on main roads such as Governor's Drive and Sta. Rosa-Tagaytay Road. Cavite Expressway and other expressways routinely have traffic bottlenecks, and what should take only a few minutes to navigate often turns into hours. For instance, in rush hour peak traffic, the commute from Cavite to Manila can exceed two hours even though the distance is approximately 30 km. In peak hours the Cavite-Manila commute often exceeds two hours, despite the roughly 30 km distance. Much of Cavite's public transit still relies on aging jeepneys, UV Express vans and provincial buses. These traditional land transport systems are congested, uncertain, and inefficient; hence, gridlock and pollution remain a challenge in the province (Highways Today, 2024; UP National Center for Transportation Studies [UP NCTS], 2025).

Trece Martires, the capital of Cavite province, plays a crucial role in the province's mobility network as it is located at the crossroads of major arterial roadways, such as Governor's Drive, which connect Dasmariñas, General Trias, and Tanza. The

Department of Public Works and Highways (DPWH) recently identified the city in an infrastructure programming plan as a point of congestion that requires road repairs and widening, indicating the increased demand for trips as well as the need to maintain an open network and access between adjacent growth centers (Department of Public Works and Highways [DPWH], 2023). Trece

Martires functions as a government and service center that attracts high volumes of trips on a regular basis which adds continued pressure to the intersections and roadside terminals. The Provincial Government of Cavite (2023) also recognized "Transportation and Traffic Management" as a major theme in its Executive-Legislative Agenda 2022-2025, denoting the acknowledgment of transportation as a basic service need in the provincial capital. However, public transport in Trece Martires remains further constrained by at-grade intersections along Governor's Drive, as well as poorly coordinated terminal planning, leading to repeated delays and unreliable service for passengers.

To address these challenges, the city government implemented its Revised City Traffic Management Code in 2022, which included rules surrounding public utility jeepneys, UV Express vans, and tricycles, changing loading and unloading practices, and traffic management (City Government of General Trias, 2022). Earlier, an ordinance prohibited tricycles from operating on national roads, as a way to support first-mile/last-mile connectivity while considering efficiency and safety issues (City Government of General Trias, 2020). Nevertheless, as is the case at various strategic intersections like Tejero Crossing, congestion remains a problem. Tejero Crossing is a critical bottleneck for intra-provincial travel for the municipalities of General Trias, Tanza, and Trece Martires. One of the measures in the Provincial Executive-Legislative Agenda (2023) points to terminal rationalization and interchange management as part of the plan for addressing congestion and reliability of transfers across modes of travel. Therefore, even when

General Trias is an example of good local government regulation for traffic and public transport operations, the city continues to contend with high volumes of vehicles, limited terminal and transfer facilities, and continued delays at key intersections, which means the efficiency and reliability of the commuters, and ultimately their travel experiences, is compromised.

In summary, the Philippines' public transport system is currently undergoing significant reforms but it remains hampered by aging, fragmented services and extreme congestion (UP National Center for Transportation Studies [UP NCTS], 2025; Gaspay et al., 2023). Cavite reflects these national issues because rapid urban growth has outstripped its transit capacity, which has led to long commutes and overcrowded informal vehicles. The government's multi-pronged modernization and infrastructure programs, which include franchising reforms, new expressways, and the planned Cavite Bus Rapid Transit (BRT) system, are designed to address these problems (Arcadis, 2023; Highways Today, 2024). If successful, these projects will not only improve the daily lives of Cavite residents through faster, safer, and more environmentally friendly commutes but will also support the larger economy by strengthening a critical growth corridor (Arcadis, 2023; Highways Today, 2024).

Multi-Modal Transportation Systems

Multi-modal commuting can be a single trip utilizing different means of travel to increase accessibility and efficiency. Multi-modal commuting can include transportation in a personal vehicle, various bus modes, biking or walking, or using micromobility (StreetLight Data, 2023). Multi-modal commuting integrates on a cohesive journey that transitions seamlessly between modes. Multi-modal transportation networks include a number of travel options and infrastructure elements that support multi-modal travel, such as safe sidewalks and crossings next to transit stops, bicycle support infrastructure, etc. Multi-modal commuting provides the four characteristics of multi-modal commuting by offering a variety of modes, sharing infrastructure elements, greater accessibility, and seamless transitions (StreetLight Data, 2023).

In the Philippines, local transport modes like jeepneys, tricycles, and multicabs will always be relevant as last-mile connections at the neighborhood level. (Viray, 2024) said planners should use the unique behaviors of jeepneys (having the ability to stop at any time) to inform simulations like PTV Vissim to the behaviours of these informal modes and formal networks. Alongside these local transport modes, we have large scale local transport modes like the point-to-point (P2P) bus system in Metro Manila that are creating a more sustainable foundation for commuters who lack private vehicles. (Cua and Wan, 2023) noted that even a 5% modal shift to P2P can reduce the CO_2 emissions by 3.2%, while the total CO_2 emissions can be approximated to a reduction of as much as 30% if there was a

57.5% modal shift. In this sense, the local modes are flexible feeders and the large-scale modes are efficient backbones, enhancing continuity, operational efficiency and urban sustainability.

Multimodal systems of transportation have valuable advantages that help improve efficiency, access, and sustainability. By linking transportation modes together - including buses, trains, bikes and shared mobility, fewer people rely on private vehicles, resulting in decreased congestion, while enabling commuters to travel seamlessly, in a reliable manner which is linked, transparent, and supported by real-time data and integrated ticketing (Chadalawada, 2022). Unfortunately, multimodal transport systems encountered challenges. In most cases, there is a gap in data for optimization, and the differences amongst geography, technology and governance prohibit predictable application of one model across the range of different contexts. In addition, a focus on urban areas over rural areas, differed levels of infrastructure maturity, and meso-level issues like last mile connectivity and congestion can set back multimodal systems capacities (Chadalawada, 2022). That being said, most of these guidance resource base their insights from literature that comes from abroad, assume strong planning frameworks, planning and physical infrastructure. In the context of the Philippines where we have informal modes of transportation like jeepneys, and little structured planning, it simply isn't possible to take these models and apply them without adaptation to local realities.

Route Optimization in Public Transportation

Route optimization in public transportation has long been studied through a range of algorithms, from traditional shortest-path models to more advanced metaheuristics. Classic approaches such as Dijkstra's and A* are efficient in identifying the shortest paths in a Point-to-Point basis, but in terms of uncertainties due to the dynamic nature of commuting they cannot really be use in that sense. Recent studies have applied to more advance metaheuristic methods such as Genetic Algorithm (GA) in the study of Bolotbekova et al. of 2025, Ant Colony Optimization (ACO), and Simulated Annealing (SA) due to these algorithms capabilities of generating a globally optimal solutions in a complex network. However, these algorithms needed iterative process for better results and this resulting to a problem with real time responsiveness and efficiency (Li, & Kim, 2024). In practice, public transport route planning involves balancing multiple factors which includes passenger waiting times, travel times, and detour lengths. This makes the problem computationally complex and beyond the reach of single-path heuristic models (Li, & Kim, 2024).

Globally, the journey planning algorithms are deployed for a real-world application. The RAPTOR algorithm for example is being implemented in the Czech Republic’s transit systems, optimizing routes based on travel time and number of transfers while offering commuters multiple alternatives within specified time ranges (Šulc & Šulcová, 2021). Although due to the limited criteria

of RAPTOR algorithm being arrival time and number of transfers. These implementations demonstrate the success of algorithm-led planning in structured transit networks that have digital schedules and formal stops. Locally within the Philippines, efforts have been limited and are in early-stage research and system development, particularly in commuter modes of public transportation like jeepneys. The iBiyahe capstone project was developed to address efficiency problems faced by commuters by developing an integrated mobile platform that used real-time jeepney tracking with live traffic updates and presented route information to users (Fraginal et al, 2024). The Study presented during this Project underwent user acceptance testing, and we were able to confirm users experienced such value through efficiency improvements and eliminating uncertainty, even if it only solves the visualization potential and not multi-criteria optimization. The project highlighted a relevant gap, as while the global systems have successfully implemented advanced algorithms with optimization, local effort is just starting to address commuters’ problems using information systems. This also shows an opportunity for the use to use both the system of algorithmic optimization and user-centric design in the Cavite context.

Route optimization is evaluated using a variety of performance metrics. Three time-based performance metrics which are; waiting time, travel time, and detour time; was mentioned in the work by Li, & Kim (2024). However, problem arises when determining if a route planning algorithm is still effectively performing when user needs are added since they need to be both balanced. In the local context

of the Philippine, a study does indicate that commuter facing performance measures matter; the iBiyahe evaluation established that live data on vehicle location and live traffic conditions enhanced efficiency and commuter satisfaction (Fraginal et al, 2024). This finding strengthens the importance of integrating not only algorithmic measures like time and cost but also reliability and usability, particularly in contexts such as Cavite where transport operations are highly variable and commuter decision-making depends on timely, accurate information.

Monte Carlo Simulation in Transportation

Monte Carlo simulation is a statistical modeling approach that embraces uncertainty by creating potential outcomes from random sampling repeated instead of relying on predetermined values. This is useful in transportation systems that utilize data based on behaviours that may vary, such as travel times, demand, service interruptions, etc. Latin Hypercube Sampling, a sophisticated form of Monte Carlo, has been used with a trip-based demand forecasting model, resulting in hundreds of parameter sets used to understand variability. The research results illustrated that a 10% shift of the assumptions made on inputs caused only about a 1% change in traffic volume forecast (Gibbons & Macfarlane, 2025) while Monte Carlo provides a realistic forecast without overestimating uncertainty.

Monte Carlo simulation is also used by transportation agencies to examine travel time reliability using a distribution of possible outcomes in place of averages. This helps to capture the variability caused by traffic congestion, route scenarios,

and variations in travel demand. One study characterized bus travel times with a distribution for buses across more than 41,000 trips and 50 routes that complementarily visualized both travel time and travel time reliability. It estimated that a similarity-based density estimation model improved upon a random forest model by nearly 9% in reducing the mean squared error of expected secondary delays (Ricard et al., 2022). Another study examined reliability in transportation networks with Monte Carlo methods, developing hundreds of scenarios under varying congestion levels. It found that variations in inputs of 10% still resulted in travel time forecasts bounded by realistic values (Liu et al., 2021).

Traffic modeling studies have also demonstrated the effectiveness of Monte Carlo approaches. (Alghamdi et al., 2022) was one of the recent studies that compared various modeling approaches, including long-term forecasting models, agent-based simulations, and stochastic approaches, and asserted that there is no silver bullet for modeling as each approach has its advantages and disadvantages with respect to accuracy, computational requirements, and flexibility. The study showed that simulation approaches, including Monte Carlo, can effectively capture the randomness in traffic flow and user behavior, and make superb research and practice tools for transportation planning. Many of the advantages and capabilities of Monte Carlo analysis are compromised in many studies that could have employed it to its fullest, even if we think of it only in terms of its specificity to reliability analysis at the experimental level. In the Philippines, it could serve a worthwhile application to really record uncertainty (e.g. unexpected stops by the jeepneys, unpredictable traffic

congestion that occurs unpredictably-degraded) if that is acknowledged explicitly in the study (Liu et al., 2021).

Genetic Algorithms in Transportation Optimization

Genetic algorithms (GA) are population-based metaheuristic search methods that evolve solutions through several key phases: population initialization, fitness evaluation, selection, crossover, and mutation. In each generation, a population of candidate routes is evaluated and the fittest solutions are combined to produce improved offspring (Liu, Zhou, Li, Wang, & Zhang, 2025; Wang, Guo, Zhang, & Zhang, 2024). Genetic algorithms perform a global search of the solution space, which helps it to circumvent local optima. GA chromosome encodings allow for different representations of a complex route, including explicit constraints such as vehicle carrying capacity or transfer functionality (Liu, Zhou, Li, Wang, & Zhang, 2025). Furthermore, GA fitness functions can account for multiple criteria fitting travel time, operating costs, and emissions (or other specified metrics) into a single objective, in the context of route planning (Wang, Guo, Zhang, & Zhang, 2024).

GAs have been applied to a number of route-planning and scheduling problems. In routing, GAs often solve vehicle-routing variants: for instance, an earlier study from Wang and Lu (2009) used a hybrid GA for the capacitated vehicle routing problem (CVRP), and reduced total travel distance by a considerable distance while considering capacity limits. Similarly, Liu et al. (2025) developed an improved GA for coordinated multi-vehicle pickup (a CVRP variant with passenger loads) that encodes routes on road segments and balances workload across vehicles. GAs have also addressed multi-modal passenger routing: Peng et al. (2021) combined GA with Monte Carlo simulation to optimize uncertain transit journeys (bus, rail, walking) under time and transfer constraints. Finally, Weng et al., (2022) used a GA to optimize city buses with respect to their timetabling problems, specifically optimizing the departure intervals to minimize passenger wait, operating costs and exhaust emissions. In each of these situations, GA is tuned search for solutions while considering realistic constraints. Notably, Bolotbekova et al. (2025) implemented a GA-based trip planner for tourist bus routes, finding that the GA outperformed other heuristics (ACO, simulated annealing, etc.) by producing the shortest feasible routes in fewer iterations. Overall, these studies indicate the extensive application of GAs for route planning, transit scheduling, and vehicle allocation.

Compared to classical algorithms, GAs trade exactness for flexibility. Traditional methods like Dijkstra’s or A* algorithm compute optimal single-criterion shortest paths very efficiently, but cannot natively handle multiple objectives or complex constraints. By contrast, GA's global search can consider multiple criteria simultaneously. For example, Liu et al. (2025) noted that GA's chromosome encoding "offers greater flexibility in encoding" and preserves feasible paths better than ant colony or PSO methods. In benchmark studies, their GA framework maintained high-quality routes under capacity and transfer constraints, whereas ant-colony-based methods struggled. Likewise, Bolotbekova et al. (2025) reported that their GA achieved the best solutions in comparison to ACO, simulated annealing (SA), and particle swarm (PSO) methods on a complex bus routing problem. Algorithms like RAPTOR (for public-transit routing) excel at fast timetable searches but assume a single cost metric (e.g. travel time). In summary, GA's multi-criteria optimization capability sets it apart: it can find solutions that balance time, cost, and other factors simultaneously. A common pitfall is to overlook this strength, since GA is often best in scenarios with conflicting objectives, whereas methods like A*, Dijkstra, ACO or RAPTOR serve single-objective queries. Studies emphasize that GA should be compared on equal footing (multi-objective vs multi-objective) to highlight its true advantages.

Recent studies from 2021 to 2025 show that genetic algorithms are effective for transportation optimization, particularly in multi-modal and multi-objective contexts (Peng, Mo, & Liu, 2021; Liu, Zhou, Li, Wang, & Zhang, 2025). Genetic algorithms have been applied successfully to route planning, transit scheduling, and various vehicle routing problems, often producing more flexible and higher-quality solutions compared to traditional single-objective methods (Bolotbekova, Hakli, & Beskirli, 2025; Liu et al., 2025). A central advantage of this approach is its ability to integrate varied criteria such as travel time, cost, emissions, and transfers within a unified search process (Wang, Guo, Zhang, & Zhang, 2024; Liu et al., 2025). The algorithm's capacity to address multiple and sometimes conflicting objectives distinguishes genetic algorithms from classical shortest-path techniques, which are limited to optimizing a single criterion.

Researchers have stressed the importance of acknowledging the distinctive capabilities of genetic algorithms while appropriately adjusting their parameters to ensure that a balance between convergence speed and solution quality can be achieved (Liu et al., 2025). Without this consideration, scholars may underestimate the amplitude of genetic algorithms for complex, transportation-related problems. If properly applied, genetic algorithms can be highly effective in optimizing scenarios that involve multiple objectives and operational constraints.

Overall, genetic algorithm-based approaches, including variants that integrate Monte Carlo simulation, are suitable for addressing the complexities of multi-modal and constrained route planning. This makes them particularly appropriate for applications such as the proposed Cavite commuter planner, where the optimization of many aspects simultaneously is essential for achieving practical and efficient solutions.

Monte Carlo–Guided Genetic Optimization (MCGA)

Monte Carlo–Genetic Algorithm (GA) hybrids have been utilized in transportation studies to model uncertainty in route planning. In a recent study, (Peng et al., 2021) used Monte Carlo simulation to model random variations or uncertainty with respect to vehicle running times and dwell times, whereas a GA was used to find the passenger paths that were shortest for multimodal transit networks while also providing reliability. The combination of the two methods allowed for a realistic representation of random behaviours while optimizing routing behaviour at the same time. The study also found that using multi-population GA approaches led to quicker convergence time and shorter CPU times, illustrating the value of the integration of stochastic simulations and evolutionary approaches for transportation modelling.

Hybrid frameworks, or a combination of Monte Carlo simulations and Genetic Algorithms (GA), have a lot of value in terms of managing uncertainty in the stochastic context and optimizing performance. For example, (Guo et al., 2024) used a hybrid approach to optimize multi-factor systems with randomness across the inputs managed by using Monte Carlo sampling and GA methods optimizing the candidate solutions toward an optimal solution. The results showed substantial improvements in the adaptability and robustness of the solution over traditional single-method approaches, and further confirmed that hybrid methods are strong tools to help solve complex optimization size in an uncertain environment.

Hybrid models that possess Monte Carlo sampling and evolutionary optimization algorithms have been successfully employed in multimodal logistics planning. (Chen et al., 2024) developed a low-carbon route optimization model for multimodal freight transport with a Monte Carlo-based catastrophe adaptive genetic sequence Algorithm to solve a problem including uncertainty about the value of cargo and delay characteristics associated with time. The model used triangular fuzzy numbers to capture variability in demand and also conducted sensitivity analyses that showed that, with some increases in carbon tax or progressively higher percentages of electric fleets, the routing decisions were unrecognizable. Their results confirm that a hybrid MC+GA framework is capable of balancing ability to manage uncertainty with a robust method for route optimization for logistics planning within a sustainable transport system. Even though these applications illustrate the robustness of MCGA in transport and logistics, its practical and complete newness as a method would be

misleading, because it has been used internationally. While underutilization of the method in local Philippine passenger transport shows the research gap, the associated lag makes exploring MCGA in the local context timely and relevant (Chen et al., 2024).

Hybrid Optimization in Multi-Modal Transportation Planning

Classical routing algorithms like Dijkstra and A* are ideal for deterministic transit networks with fixed period and routes. However, they do not capture stochastic variability which characterizes informal public transportation systems, in which vehicles provide service at different frequencies, fluctuate unpredictably. Single-objective heuristics similarly do not provide a way of representing the trade-offs commuters undertake between, e.g., time of travel, cost of travel, environmental impact, or transfer frequency.

Monte Carlo (MC) simulation, in contrast, overcomes these limitations by utilizing repeated stochastic sampling of a variable, such as distributions of travel time or varying vehicle arrivals, by modeling uncertainty of variables. Within transportation research, MC methods have been employed to assess public bus travel time reliability and assess network disruption impacts (Liu, Wu, Liu, Sun, & Wang, 2021; Ricard, Desaulniers, Lodi, & Rousseau, 2022).

Genetic Algorithms (GA), as previously noted, handle multiple conflicting objectives and can traverse large solutions spaces. When using GAs in stochastic settings, as a singular instance of optimization, it may fail to capture real-world

variation behavior resulting in solutions for a system whose performance will vary under different conditions.

The Hybrid Monte Carlo and Genetic Algorithm (MCGA) frameworks combine both approaches. MC is utilized to measure uncertainty in variables which include variables such as jeepney headways, traffic conditions, and GA is utilized to improve multi-criteria routing decisions with uncertainty in in the two systems (transportation applications). Recent studies found strong evidence that MCGA has many advantages over using either standalone methods (Peng, Mo, & Liu, 2021; Chen, Hu, & Liu, 2024; Guo, 2024).

The informal public transportation context of Cavite (where jeepneys, tricycles, and multicabs operate without a clearly defined schedule) represents a context in which MCGA offers unique advantages over other optimization techniques. Traditional algorithms make assumptions of stochastic conditions and fail when considering variations in travel times. Other approaches, like RAPTOR, assume a formalized timetable of services is necessary. In contrast, MCGA can optimize multiple objectives simultaneously (travel time, cost, emissions, waiting time, transfers) and incorporate stochasticity in vehicle availability and travel congestions.

While International studies have documented the use of MCGA in formalized transit networks, however there are no documented uses of the MCGA in an informal paratransit system in the Philippines. This study addresses this gap by applying the MCGA for mode routing in Cavite.

Table 1

Comparative Analysis of Different Algorithms

Criteria	**A***	**RAPTOR**	**Genetic Algorithm (GA)**	**Monte Carlo Simulation (Standalone)**	**MCGA (Proposed)**
Convergence Speed	Faster than Dijkstra on grids	Fast; processes routes in rounds	Moderate; slower than gradient methods	Not used for optimization	Moderate; slower than A* but finds robust solutions
Solution Quality	Optimal with admissible heuristic	Pareto-optimal for time and transfers	Near-optimal for complex problems	N/A	Near-optimal across stochastic scenarios; robust to variability

Scalability	Efficient on small-moderate grids; weakens on large	Highly scalable; handles millions of departure events	Handles medium to large problems	N/A	Handles medium to large problems with stochastic inputs
Handling Discrete Variables	Designed for grid-based discrete pathfinding	Excellent for discrete transit stops and routes	Well-suited for discrete and nonlinear problems	Models' variability, not discrete optimization	Excellent; combines GA's discrete handling with MC's uncertainty modeling
Robustness to Local Minima	Avoids local minima with consistent heuristics	Not applicable; deterministic algorithm	Good at avoiding local traps	N/A	High; GA explores solution space while MC evaluates robustness
Computational Complexity	40% more efficient than alternatives on grids	Low; computes journeys in under 6 ms for large networks	Moderate to high, depends on population size	Depends on number of runs)	High; requires multiple MC runs per GA generation
Practical Application	Widely used despite niche limitations	Public transit journey planning	Widely used in transport and robotics	Used for uncertainty modeling	Emerging in stochastic transport and logistics optimization

Suitability for Multi-Modal Transport	Limited without multi-agent extensions	Limited; requires fixed timetables	Highly suitable for mode switching	Supports stochastic modeling	Highly suitable; optimizes multi-modal routes under stochastic conditions
Handles Uncertainty	Assumes deterministic costs; heuristic-limited	No; requires deterministic schedules	Limited	Yes; quantifies stochastic variability	Yes; explicitly models and optimizes under uncertainty

Table 1 shows a comparison of the routing algorithms on the basis of the nine criteria that are most influential for multi-modal transport. The A* algorithm is the best solution for the quickest, most optimal grid path, but its ability to process large data sets is limited and it requires predetermined costs (Foead et al., 2021). The RAPTOR algorithm is the most advanced among the transit schedule-based routing algorithms as it deals with very large transit schedules with ease but needs fixed time tables. Genetic Algorithms are used to solve the most complex, intricate, and disconnected problems robustly (Doroshenko, 2025; Wang et al., 2024). Monte Carlo is a method that models uncertainty without the previous optimization step being applied (Peng et al., 2021). The hybrid MCGA is the method that combines GA exploration with Monte Carlo evaluation for robust stochastic route optimization.

Carbon Footprint in Transportation Systems

Transportation is one of the significant contributors to carbon dioxide emissions (Yaacob et al. 2020), largely due to the dependence on fossil fuels. In the study of Vergel et al. (2024) provides recent estimates of local pollutant and CO_2 emissions for road and rail transport in the Philippines, showing the scale of transport's environmental impact. Their work highlights how road-based modes contribute disproportionately compared to rail, underscoring the need for accurate local emission baselines. These findings are supported by global reviews which show that vehicle type, fuel consumption, speed, and weight are variables to consider when looking at emission levels (Yaacob et al. 2020). This suggests that emission factors specific to jeepneys, tricycles, and multicabs are crucial for designing sustainable route optimization systems in Cavite.

Transport contributes more to air pollution in terms of constantly emitting CO_2 (Yaacob et al., 2020), calling the need for commuting strategies. This is in line with the United Nations' Sustainable Development Goal #13 (Climate Action), which emphasizes the need for urgent action to reduce greenhouse gas emissions. In the Philippines, reports on transport emissions are increasing, and while there are some examples of low carbon emission strategies, most do not target a shift or commence with commuter behavior but instead they target modernization of the public utility vehicles (Vergel et al., 2024). Commuters could also consider the carbon footprint of their commuting decisions in regards travel which contributes to sustainability targets.

Ultimately, linking people's travel behaviors to larger environmental metrics provides not only an academic rationale but also a societal justification for carbon-relevant systems of public transportation.

International research has already shown that carbon-aware route optimization can be effective. Zrigui et al. (2023) highlighted that real-time transport planning with big data analytics can reduce fuel consumption and CO_2 emissions. Moreover, the simultaneous focus on reduced costs and sustainability demonstrates that eco-routing can be practical. The use of carbon metrics to assess the “eco-friendliness” of the route can be a good integration in a journey planning app. Although Google Maps mentioned the eco-friendly routes being a feature it is only available in selected countries and regions (Use Eco-friendly Routes on Your Google Maps App - Android - Google Maps Help, n.d.). This gap demonstrates the novelty of a Cavite-focused eco-aware routing system that utilizes jeepneys, tricycles and multicabs.

Web & Mobile-Based Transit Applications

Jeepneys, are the principal and cheap public transport in the Philippines, despite the long-standing issues concerning safety, maintenance, environmental suitability, and a lack of formalized loading and unloading (Andalecio et al., 2020). These issues affect commuters themselves, but they additionally create challenges in integrating jeepneys/tricycles into global transit apps that depend on structured datasets like structured datasets, fixed routes, and official timetables. Google Transit states that their partnership program seeks static data from their partners such as detailed transit stops but informal transportation modes like the jeepney may not have well-defined stops (Google Transit Partners, n.d.).

The Transformative Urban Mobility Initiative (TUMI) provides General Transit Feed Specification (GTFS) data for Manila through collaboration with agencies such as the LTFRB, LRT Authority, and Philippine National Railways. This dataset, however, is timetable-bound and limited to formalized systems such as rail and bus rapid transit. Informal transport modes such as jeepneys, tricycles, and multicabs which are dominant mode of transportations in Cavite and other peri-urban areas are excluded. In addition, because General Transit Feed Specification (GTFS) data may become outdated over time, its usefulness for local commuters who depend on real-time and highly variable modes of transport is limited.

Commuter expectations for transit applications are not only speed and accuracy but also usability, offline access, and adaptability to local travel conditions (Strenitzerová & Stalmachová, 2021). This shows that although Functional

Requirements or the features builds the app the Non-Functional Requirements should not be overlook.

Thus, while databases like General Transit Feed Specification-Manila, and global apps like Google Transit are valuable in terms of formalized urban systems, they cannot offer the same features when it comes Cavite's transportation. A localized application with a focus on commuters' that digitizes the informal routes aspects, along with usability, flexibility, and sustainability, is needed to supplement the formal apps and provide value to both commuters and operators.

Table 2

Comparative Analysis of Existing Apps

App Name	Description	Relevance /Similarity	Uniqueness of the Proposed System
Google Maps Transit	Google's public transit service showing routes and modes of	Similar in offering commuter route options. But limited to selected modes of	Incorporates Cavite's informal modes (jeepneys, tricycles, multicabs), which lack

	transportation.	transportation such as Bus and Trains since it requires a dataset of locations and details of transit stops	standardized datasets and detailed stops.
Sakay.ph	Local app that provides routes and fare estimates for jeepneys, buses, MRT, and LRT in Metro Manila	Similar to study goals of commuter-facing tools. Breaking down each mode of transportation to use	Localized for Cavite's transport network and integrates optimization models that account for uncertain travel times, multiple criteria, and eco-aware routing.
Moovit	A Global Commuter App that is a Mobility as a Service solutions Provider	Comparable in combining formal transport and commuter data	Uses government-supported local datasets for greater accuracy in Cavite's transport conditions.
Grab/Angkas/Move it/JoyRide	Local motorcycle taxi apps with real-time booking and tracking.	Similar in providing commuter mobility solutions.	Provides multi-modal route planning that integrates various modes rather

		than single point-to-point rides.

Table 2 comparison shows that although existing applications provide useful features, it is most likely in a specific and cannot fully address Cavite's fragmented transportation system. The present study responds to these gaps by developing a localized, commuter-focused route planner that integrates informal modes, optimization models, and eco-aware routing.

Software Quality Standards for Transit Systems

Software quality in multi-modal transit applications is often assessed using established models such as ISO/IEC 25010, which defines product-quality attributes including functional suitability, reliability, usability, and performance efficiency (Maia et al., 2021). These dimensions are critical for commuter systems since applications

must provide accurate routes, handle high request volumes without crashing, remain intuitive for everyday users, and deliver results quickly in real time. For instance, usability, reliability, and performance efficiency have been repeatedly identified as essential indicators of mobile application quality (Maia et al., 2021). In the context of route planning, functionality ensures the accuracy of itineraries, reliability guarantees system availability during demand surges, and efficiency emphasizes fast route computation that is crucial in real-time decision-making.

In practice, researchers rely on user-acceptance frameworks and usability metrics to evaluate these systems. The Technology Acceptance Model (TAM) is commonly used to establish the link between perceived usefulness, usability, and user satisfaction. Moreover, standardized tools such as the System Usability Scale (SUS) provide structured ways to measure ease-of-use through user surveys. For example, Putra et al. (2024) reported a mean SUS score of 75 for a travel application, which was interpreted as a “Good” level of usability, though improvements were still recommended in terms of interface responsiveness and navigation clarity. In addition to surveys, task-based metrics such as average completion time for trip planning or error rates during navigation tasks are frequently used to evaluate commuter-facing systems (Brata et al., 2021). These frameworks and tools provide tangible evidence to assess how well an application aligns with ISO/IEC 25010 usability and efficiency requirements.

Empirical studies demonstrate the value of applying software quality standards in commuter-oriented applications. Cabatit et al. (2023) assessed “Viajeros,” an online minibus reservation system, and found that commuters perceived the software to be highly functional, reliable, and usable when evaluated against ISO quality attributes. Similarly, Pizarro et al. (2025) evaluated the TC-TRISSEA tricycle commuter application in Tuguegarao City, where IT experts rated the system as compliant “to a very great extent” with ISO/IEC 25010 criteria. These evaluations highlighted the significance of aligning transit systems with functionality, reliability, and usability standards to ensure commuter trust and satisfaction. Performance efficiency also emerged as a recurring theme, as both systems emphasized the importance of responsiveness to real-time conditions. This finding is particularly relevant for Cavite, where traffic congestion and variable commuting conditions necessitate fast, reliable, and accurate route planning.

Literature shows that ISO/IEC 25010's attributes of functionality, usability, reliability, and performance efficiency are central in the evaluation of commuter-focused transit systems (Maia et al., 2021). Frameworks such as TAM and usability instruments like SUS, combined with metrics such as completion times, offer systematic ways to measure acceptability and performance (Brata et al., 2021; Putra et al., 2024). Case studies demonstrate that when applications are evaluated against these standards, they achieve higher usability ratings and commuter satisfaction (Cabatit et al., 2023; Pizarro et al., 2025). To address practical evaluation, usability can be measured with SUS surveys, while efficiency can be assessed by monitoring response times and task completion speed. Together, these approaches ensure that multi-modal transit systems meet the demands of real-time commuting in Cavite.

Summary & Research Gaps

The review of related literature indicates that public transportation in the Philippines continues to suffer from historical issues such as congestion, aging vehicles, and lack of organization. As the most accessible and affordable modes of transportation, jeepneys and tricycles are burdened by issues of safety, unpredictability, and negative environmental impacts (UP NCTS, 2025; Andalecio et al., 2020). In Cavite, these concerns are made worse by extensive urban growth and a lack of transport infrastructure, generating long journeys and prolonged travel times, unorganized loading/unloading, and unreliable travel times (PSA CALABARZON, 2023; Lara & Perido, 2024). Modernization and infrastructure programs are in place but the preexisting system is still not correlatively able to satisfy commuters' demand.

Globally, Multi-modal transportation systems emphasize that integrating modes of travel can provide seamless and efficient travel experiences (StreetLight Data, 2023; Chadalawada, 2022). But most of these systems assume formalized routes and digital schedules and have strong institutional backing. Given the informality of much of the transport system in the Philippines and limited data on routes and operations, these transportation systems are difficult to apply directly to the Philippines. Nevertheless, it is shown that optimizing routes through approaches like Genetic Algorithms (GA), Ant Colony Optimization (ACO) and RAPTOR can improve travel efficiencies and lower operational costs in formalized contexts (Li & Kim, 2024; Šulc & Šulcová, 2021).

Furthermore, optimization methods tend to focus less on commuter-centered concerns like accessibility, waiting time, and reliability, especially in local contexts of Cavite.

Monte Carlo simulations have been adopted to represent uncertainty and take into account various unexpected changes in transport systems, such as random delays or changes in traffic flow (Gibbons & Macfarlane, 2025; Liu et al., 2021). The work done in combining Monte Carlo with GA suggests how hybrid methods can provide a more reliable and adaptable outcome (Peng et al., 2021; Chen et al., 2024). Despite this potential, these methods are still largely absent in local public transport optimization research, creating an opportunity for research.

Another limitation that recurs is the failure of digitalization through web and mobile applications of informal modes, such as jeepneys and tricycles. Platforms such as Google Maps often rely on structured datasets, which cannot transfer the agitated nature of these modes (Google Transit Partners, n.d.; TUMI, 2024). Commuter-centered research has also indicated that user expect not just accurate directions, but functional usability, accessibility without internet, adaptable to local conditions (Strenitzerová & Stalmachová, 2021). The reach of current apps suffers user expectations, particularly in peri-urban areas like Cavite.

In summary, the literature points to several key research gaps: (1) the limited application of hybrid optimization and simulation methods in local transport systems, (2) the exclusion of informal modes from existing digital platforms, and (3) the absence of commuter-centered design that accounts for local realities. These gaps highlight the need for a localized, data-driven, and commuter-oriented system that integrates multi-modal transport, optimization algorithms, and adaptive features to improve the reliability and sustainability of commuting in Cavite.

Conceptual Framework

Figure 1

Conceptual Framework of NaviCav

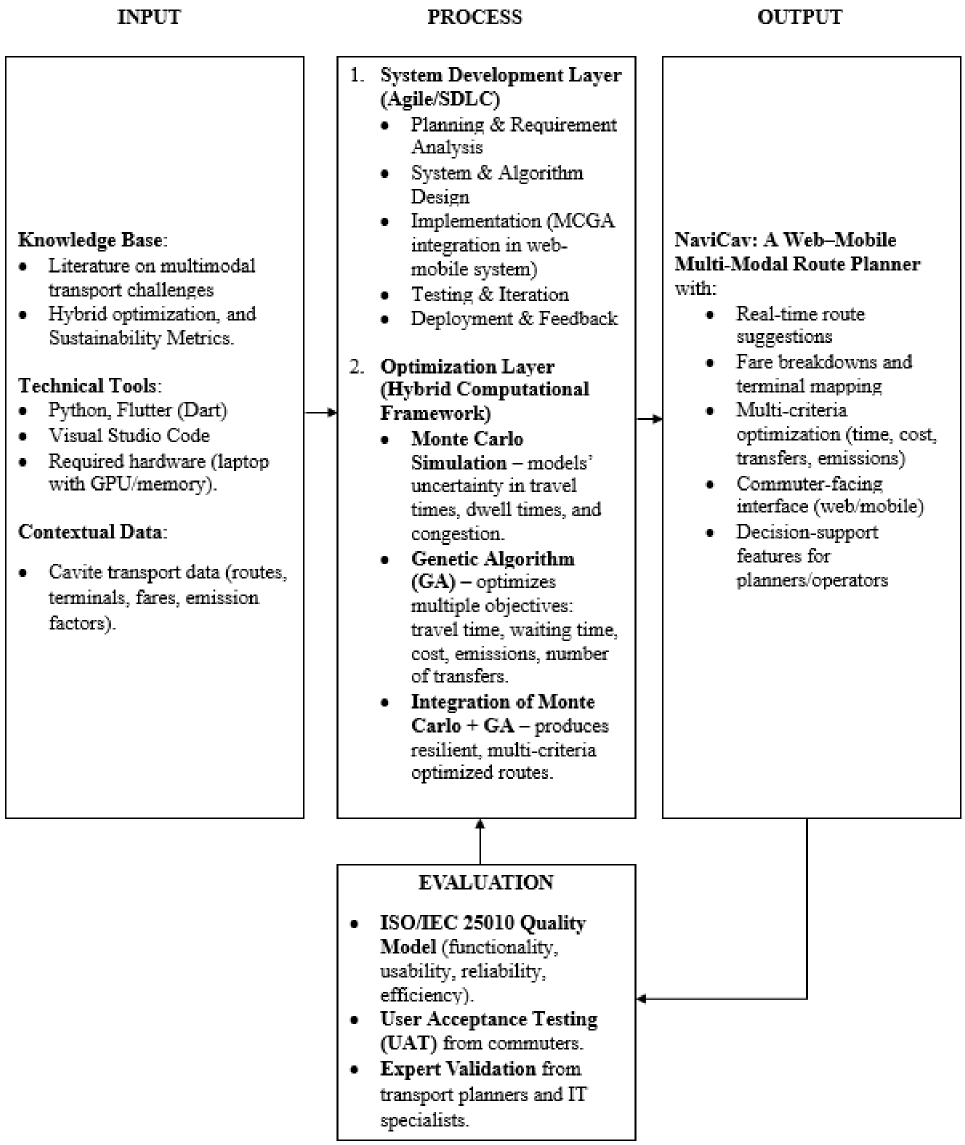


Figure 1 shows the conceptual framework of the study which explains the Input–Process–Output (IPO) model with evaluation. The Input stage outlines the requirements needed to initiate the project, including knowledge base of relevant literature, hybrid optimizations, and sustainability metrics. For the Technical Tools programming languages (Python for backend, Dart which is used on flutter framework). Software requirements, such as Visual Studio Code and Windows OS, alongside hardware requirements like a laptop with sufficient memory, GPU, and storage, provide the technological foundation for development. Lastly is the contextual data Cavite Transport Routes specifically the locale of the study which are the municipalities of General Trias, and Trece Martires and for that we'll be using the official city government document which will be the Local Public Transport Route Plan (LPTRP).

The Process stage comprises two layers. The first layer, System Development Layer, follows Agile methodology comprising planning, requirement analysis, system and algorithm design, implementation, iterative testing, and feedback, and deployment. The second layer, Optimization Layer, is a hybrid computational framework in which Monte Carlo simulations model uncertainties in travel times and congestion and Genetic Algorithm optimizes multiple objectives including travel time, cost, emissions, waiting time, and number of transfers. The use of Monte Carlo and GA produces resilient multi-criteria optimized routes in the transport context of Cavite.

The Output of the process is NaviCav, a web–mobile multi-modal route planner for Cavite. It provides real-time route suggestions, fare breakdowns, and terminal mapping. The system applies multi-criteria optimization to deliver possible routes that balance travel time, cost, number of transfers, and emissions. It includes a commuter-facing interface for everyday users and decision-support features for transport planners, enabling practical and efficient route planning across multiple modes of transport.

The Evaluation stage measures the application's quality using the ISO/IEC 25010 standards, assessing its functionality, usability, reliability, and efficiency. Together with User Acceptance Testing (UAT), commuters ensure the system meets their needs, while expert validation from transport planners and IT specialists confirms that it satisfies technical and operational requirements.

Definition of Terms

For the readers to fully comprehend this study, the following are the notable terms that have been utilized in the study:

Public Utility Jeepney (PUJ/Jeepney). A traditional and widely used mode of public transport in the Philippines, typically characterized by fixed routes but informal loading and unloading practices.

UV Express. A point-to-point utility van service in the Philippines providing faster intercity connections than jeepneys.

Tricycle. A motorcycle with an attached sidecar used as a local public transport mode for short distances.

Multi-Modal Transportation. A system of travel where commuters use two or more modes of transport (e.g., jeepney + bus, or walking + train) within a single journey.

Point-to-Point (P2P) Bus. A bus service with limited, direct stops intended to reduce travel time between major origins and destinations.

Public Utility Vehicle Modernization Program (PUVMP). A government initiative aimed at replacing traditional jeepneys with modernized vehicles to improve safety, environmental performance, and efficiency.

Route Optimization. The process of determining the most efficient path between origins and destinations, considering criteria like travel time, cost, and number of transfers.

Shortest Path Algorithms (Dijkstra's, A*). Classical graph algorithms used to find the minimum-distance or minimum-cost path between two nodes in a network.

Metaheuristic Algorithms. Advanced optimization approaches such as Genetic Algorithm (GA), Ant Colony Optimization (ACO), and Simulated Annealing (SA) designed to find near-optimal solutions in complex search spaces.

Genetic Algorithm (GA). A population-based optimization algorithm inspired by biological evolution, using selection, crossover, and mutation to evolve solutions.

Ant Colony Optimization (ACO). A probabilistic optimization algorithm inspired by the behavior of ants searching for food, often used in routing and scheduling.

Simulated Annealing (SA). A probabilistic technique for approximating the global optimum of a function, modeled after the process of heating and cooling metal.

RAPTOR (Round-Based Public Transit Optimized Router). A public transport journey planning algorithm that optimizes travel based on arrival time and number of transfers.

Monte Carlo Simulation (MC). A computational method that uses random sampling to simulate variability and uncertainty in outcomes such as travel time or demand.

Monte Carlo–Genetic Algorithm (MCGA). A hybrid optimization technique combining Monte Carlo simulation's capacity for uncertainty modeling with GA's optimization strength.

Performance Metrics. Criteria used to evaluate route optimization, commonly including waiting time, travel time, detour length, reliability, and cost efficiency.

GTFS (General Transit Feed Specification). A standardized data format used by transit agencies worldwide to provide static transit information such as stops, routes, and schedules.

GTFS-Manila. The GTFS dataset curated by the Transformative Urban Mobility Initiative (TUMI) covering Metro Manila's formal transit modes such as rail, BRT, and buses.

Google Transit Partners Program. A program by Google requiring structured static data (e.g., stop locations, routes) from transport agencies in order to integrate their services into Google Maps.

Eco-Route Planning. A transport planning approach that incorporates environmental impacts, such as fuel use and CO_2 emissions, into route optimization.

Carbon Footprint. The total greenhouse gas emissions produced directly or indirectly by transportation activities, typically measured in CO_2 equivalents.

ISO/IEC 25010. An international standard defining quality attributes of software products, including functionality, reliability, usability, and performance efficiency.

System Usability Scale (SUS). A widely used questionnaire for assessing user perceptions of software usability.

Technology Acceptance Model (TAM). A framework used to evaluate user acceptance of technology, focusing on perceived usefulness and ease of use.

Colorum. Public Utility Vehicle (PUV) that operates without the proper permits or official documents.

CHAPTER III

METHODOLOGY

Research Design

NaviCav's research approach will be hybrid and will rely on both quantitative and qualitative approaches for a full examination of the system. The qualitative part will focus on exploring users' subjective evaluations through surveys, particularly usability and overall experience as commuters. The information will be helpful in explaining how the app will cater to public transportation users specifically. The quantitative dimension, on the other hand, will objectively evaluate whether the Monte Carlo–Guided Genetic Algorithm (MCGA) will perform within the context of multi-modal transportation in the Philippines, where a significant portion of the informal transport modes are jeepneys and tricycles, among others. The hybrid approach will ensure that both user experience and algorithmic validity are consistently investigated.

The selection of the MCGA addresses unique challenges of informal public transit in Cavite. Classical deterministic algorithms like Dijkstra and A* algorithms assume fixed edge weights and cannot model the stochastic variability of jeepney headways, tricycle availability, and traffic congestion (Rachmawati & Gustin, 2020). Schedule-based algorithms like RAPTOR require formalized schedules that do not exist for informal paratransit modes (Šulc & Šulcová, 2021). Standalone Genetic Algorithms optimize multiple objectives, yet lack mechanisms for quantifying the input uncertainty (Bolotbekova et al., 2025), while Monte Carlo quantifies the uncertainty but does not optimize route selection (Peng et al., 2021).

MCGA fused both approaches by using Monte Carlo simulation to produce probabilistic distributions of travel times, costs, and emissions based on stochastic scenarios, while the Genetic Algorithm evaluates and selects routes that remain robust under uncertain conditions (Peng et al., 2021; Chen et al., 2024; Guo, 2024). This will enable NaviCav to recommend routes that balance five competing objectives: travel time, cost, emissions, waiting time, and transfer count while accounting for the inherent unpredictability in Cavite's informal transport network.

Dataset Description and Processing

The primary data source for this study is the Local Public Transport Route Plan (LPTRP), a document issued by the local government unit (LGU) with a typical validity of 10 years. The document contains important geographic and transport information including city locations, coverage areas, major roads, routes, and terminal points. Additional fare data will be obtained from fare matrices requested from the City Government and, as a reference, the Jeepney Fare Matrix published by the Land Transportation Franchising and Regulatory Board (LTFRB) in 2023.

The raw data includes mostly PDFs and scanned documents that need to be digitized manually. Route data will be traced using UMAPS, a geospatial mapping tool, to accurately draw the routes and export them into GeoJSON format for the systems to use.

The set of procedures for the preprocessing of the data includes:

- Digitizing transport routes and terminals into GeoJSON files through UMAPS process
- Standardizing route names, terminal points, and geographic coordinates
- Handling missing or unclear sections through consultative validation with LGU departments, such as the Traffic Management Office
- Adding updates or new LPTRP versions as they are available, to maintain currency of data

Since the LGU is currently developing updates and revisions to the LPTRP, it is likely that there will be variability in data quality across LPTRP versions. Strong validation processes will be put in place, in partnership with the appropriate City department, to confirm accuracy and completeness of the digitized data, and validation processes will include cross-checking against the formal documents, as well as against transport management expert feedback.

The mixed-source and partially manual processing approach will ensure that the data used in the NaviCav system is as accurate, complete, and up to date as possible for use in the multi-modal route planning algorithm.

To ensure the accuracy of the routes and stops that will be digitized manually, geospatial validation will take place. Each of the identified routes from the LPTRP PDF Documents will be cross-checked with the satellite images available through OpenStreetMap to ensure that the digitization properly reflects the actual road networks. Where possible, GPS traces from initial fieldwork (and in subsequent processes) will be utilized as source data to substantiate the route geometry. If there are differences greater than 50 meters between the digitalized routes and the satellite images, the data will be reviewed and adjusted manually for accuracy.

For incomplete fare data, interpolation will be conducted utilizing the official jeepney fare matrix from the Land Transportation Franchising and Regulatory Board (LTFRB) website as the default reference. Distance-based fare matrices, available from the LPTRPs, will act as supporting reference based on these routes. Important to note, if there is any conflicting or ambiguous information regarding the fare, the source of truth will be the Local Government Unit Transportation Office. Any price values will also be confirmed and verified against applicable Municipal Ordinances regarding public transport fare policy.

The project database will handle all datasets processed for the project, which contains routes, stops, fares, and transit information needed for NaviCav's routing capabilities. The raw LPTRP documentation acquired from the LGU will be stored securely with access restricted to the research team and will not be posted in publicly available repositories because of privacy and access-related issues. This ensures data security while still allowing for verification and updates from source documents.

Sampling Technique

A mixed-methods study will perform the assessment of the NaviCav system along with its MCGA optimization engine using a hybrid sampling approach. Convenience sampling will be done to pick public transport passengers from Trece Martires and General Trias. This will be the case because it is a practical and feasible means of getting the data from the people who are nearest, especially the daily commuters and local residents, who, in fact, suffer the transport problems the study wants to solve. Besides, convenience sampling will also be appropriate as the user attitudes and usability insights are best obtained from people who use multi-modal transport in the study area frequently and are thus regular commuters.

Technical performance of the system will be measured through the selection of technical experts by purposive sampling. The experts will comprise planners and IT professionals in transportation, who are knowledgeable enough in the domain to perform an assessment of the NaviCav platform and the Monte Carlo-Guided Genetic Algorithm. The use of purposive sampling will be justified because the evaluation calls for the involvement of professionals with specialized expertise who will render feedback that is informed about the system's functionality, the performance of algorithms, and the applicability of industrial practices.

The algorithmic testing sample will comprise the complete dataset of transport route data from Trece Martires and General Trias. The dataset will cover the major corridors, terminals, and transfer points, which means the MCGA will be tested under the multimodal transport conditions that are realistic considering the travel time variations and actual commuter movement patterns.

Participants of the Study

The research will involve two main groups: local commuters from the locale and transport planning and management experts. Commuters will be defined as persons who regularly use phased or multi-mode public transportation options, including jeepneys, tricycles, and buses geared toward passenger use. Experts in public transportation systems will include city leaders, urban planners, or transportation professionals who are relevant to the area.

The participants who will be selected for the study will be relevant and diverse in both demographic backgrounds (age, frequency of commuting, and use of technology applications for navigation purposes), as well as in their daily use of transportation. The criterion for commuter participants will be defined by those who use multi-mode public transportation within the locale. The criteria for expert participants, on the other hand, will be determined by their knowledge of transport planning or involvement in policymaking. Any individual who will not have relevant commuting experience or appropriate professional involvement will be excluded to maintain data validity and reliability.

Table 3

Frequency of the Participants of the Study based on Category

Participant Group	Inclusion Criteria	Sample Size
Local Commuters	Must use multi-modal public transport within Trece Martires City and General Trias City; diverse in age, commuting frequency, and technology use for navigation	100
Transport Specialists	Individuals with expertise in public transport operations, mobility systems, or transport network analysis within the local or regional context.	5
Urban Planners	Professionals involved in land use planning, city development, or transportation planning relevant to urban mobility in Cavite.	5
IT Experts	Professionals with experience in software development, system evaluation, or technical design aligned with routing, optimization, or mobile–web system development.	

According to Table 3, local commuters will make up the majority of the planned participants (n = 100, 86.9 percent) while a smaller but well-defined group will be made up of transport specialists, urban planners and IT experts (n = 15, 13.1 percent). Local commuters will be the main users of the NaviCav platform and will be the ones providing the most important feedback concerning daily travel behavior, system usability, and multi-modal commuting conditions. On the other hand, the three technical expert groups will offer professional evaluations that will influence the assessment of route accuracy, optimization performance, and overall NaviCav feasibility in the local mobility context. The bigger commuter group will guarantee that different user profiles are rightly represented and at the same time, the experts will provide technical rigor necessary for evaluating system functionality and design.

Participants and Roles

a. **Local Commuters (100 participants).** The NaviCav system will get its primary end-users from local commuters. The user group will include persons who are daily users of different public transportation modes in Trece Martires City and General Trias City. The inquiries from the users will be supporting assessment of the road clarity, handling quality, perceived travel-time accuracy, and the system's reactiveness in providing real-world commuting needs. Additionally, the passengers will contribute data on travel patterns, both peak and off-peak, and hills, steps, and so on that need to be taken when changing from one transport mode to another like jeepneys, tricycles, and buses. With this wide sample size, the study will be able to reflect the whole spectrum of demographic and behavioral differences, along with varying levels of familiarity with navigation technologies.

b. **Transport Specialists (5 participants).** The group of transport specialists will be made up of professionals who have knowledge in the areas of public transport operations, mobility analysis, or transportation systems within the local or regional context. The specialists will provide the necessary input for the evaluation of the routing logic and multi-modal integration used in the NaviCav platform, whether the routing logic and multi-modal integration really are with transport operations. Their involvement will be very important for judging the possibility of route outputs and how consistent the platform is with the prevailing mobility patterns and operational constraints.

c. **Urban Planners (5 participants).** Professional perspectives in connection to land usage, road network and urban transportation will be contributed by urban planners. Their participation will facilitate the evaluation of the system's compatibility with the city planning objectives, infrastructure constraints, and long-term mobility strategies in Cavite, as such the evaluation made by the urban planners will be crucial to the decision-making on the appropriateness of the platform's outputs and spatial representation for real-world deployment.

d. **IT Experts (5 participants).** The professionals in IT will be issuing technical evaluations in connection with software quality, system capacity, interface friendly use, and system performance. The input from IT experts will assist in deciding if NaviCav platform satisfies the technical requirements of contemporary web-mobile apps. The experts will be assessing both the functional and non-functional parts of the system, examining the integration of the Monte Carlo-Guided Genetic Optimization approach, and pointing out the aspects where system dependability and user-friendliness can be enhanced.

Research Locale

Geographic Scope. The study area comprises two component cities in the province of Cavite: Trece Martires City and General Trias City. All primary data collection, network mapping, simulation parameterization, and evaluation activities will be limited to these two cities and the principal arterial corridors and terminals that connect them to adjacent municipalities and regional expressways. The research will therefore focus on origin–destination pairs, terminals, transfer points, and feeder links within the urban and peri-urban extents of Trece Martires and General Trias.

- Relevance to study objectives. Both cities exhibit concentrated commuter demand and the modal mix (jeepneys, UV Express, buses, tricycles) that the MCGA framework is intended to represent and optimize (UP NCTS, 2025; Zafra, Briones, & Alampay, 2023).
- Complementary urban profiles. Trece Martires functions as an administrative and service center while General Trias shows rapid residential and commercial growth; together they expose the algorithm to differing land-use patterns, peak directions, and modal interactions, improving external validity of results (General Trias City Government, 2020).
- Corridor significance. Major corridors and intercity connectors in this subregion of Cavite experience recurring congestion and serve as principal commuter routes, enabling realistic calibration of stochastic travel times and transfer

penalties used in simulation and optimization (Department of Public Works and Highways, 2023; Provincial Government of Cavite, 2023).

Demography and Growth. According to the Philippine Statistics Authority CALABARZON (2021), Cavite remains one of the most densely populated provinces in the country, with Trece Martires and General Trias among the leading contributors to its rapid population growth. Trece Martires recorded a population of 210,503 in the 2020 census, while General Trias reported 450,583 residents (PSA CALABARZON, 2021). These figures illustrate the strong commuter activity and mobility demands across the two cities, both of which are linked through expanding residential, educational, and commercial zones. The growing population intensifies the need for efficient transport planning and multimodal optimization, which this study seeks to address.

Transport Network and Modal Mix. Both cities are served predominantly by road-based public transport. The modal mix includes intercity and city jeepneys, UV Express vans, provincial and point-to-point buses, and tricycles as first-mile/last-mile feeders. Terminals, municipal markets, schools, and commercial centers function as major trip generators and transfer nodes. The absence of local passenger rail service within the two cities focuses the study on road-based multimodal interactions, transfers at terminals, and stochastic variability in headways and dwell times (Zafra et al., 2023; Highways Today, 2024).

Infrastructure and Corridors. Key arterial corridors that traverse or bound the study area are subject to infrastructure works and traffic management programs identified in regional plans and DPWH programming documents. Notable corridors such as Governor's Drive, Arnaldo Highway, and the Tanza–

Trece Martires–Naic Road are under expansion or rehabilitation to accommodate increasing vehicle volumes (DPWH, 2023; Provincial Government of Cavite, 2023). These developments not only enhance road connectivity but also present a realistic test environment for modeling multimodal commuter routes under variable traffic conditions. Public facilities such as city halls, markets, schools, and business districts generate consistent commuter flows that further contribute to dynamic travel patterns, which are central to the MCGA model's calibration.

Access, Permissions, and Coordination with LGUs. Prior to data collection, the researchers formally coordinated with the local government units (LGUs) of Trece Martires City and General Trias City. Official permissions and endorsements have already been secured from both the City Mayor's Offices and the City Planning and Development Offices. These permissions include authorization for data collection, access to relevant transport and demographic information, and coordination with barangay officials for local fieldwork activities.

In compliance with ethical research standards, the researchers submitted an official research proposal, an ethics and data-use statement, and detailed fieldwork plans outlining data-gathering procedures. Barangay-level coordination has also been established to facilitate on-site surveys and observational studies. With these permissions in place, all subsequent fieldwork and data-gathering activities are conducted in accordance with local regulations and institutional research protocols.

Rationale and Suitability of the Research Locale. Trece Martires City and General Trias City were selected as the focal areas of this study because of their strategic relevance to commuter mobility and transport network optimization in southern Cavite. Both cities offer distinct yet interconnected urban conditions that align with the objectives of the MCGA framework. Trece Martires serves as the administrative core of the province, hosting government offices and provincial services that attract daily commuter flows, while General Trias continues to expand rapidly through residential and industrial developments that generate significant transport activity (General Trias City Government, 2020). The interplay between institutional concentration and urban expansion provides an ideal setting for evaluating the MCGA model across varied traffic and route conditions.

The cities' transport systems are characterized by complex interactions among multiple public transport modes, including jeepneys, UV Express vans, tricycles, and provincial buses. These modes contribute to dynamic travel behaviors, offering a practical context for testing optimization under multimodal conditions (Zafra et al., 2023). Ongoing infrastructure improvements in both cities, as indicated by the Department of Public Works and Highways (2023) and the Provincial Government of Cavite (2023), further enhance their suitability as research locales by providing updated and evolving road environments for empirical assessment.

Having secured permissions from local authorities, the researchers are fully authorized to conduct surveys, route inventories, and observational studies within both cities. This collaboration ensures logistical support and access to relevant transport data, reinforcing the reliability and integrity of the study's data-gathering process. The established coordination also facilitates adherence to local policies, ensuring that all activities are aligned with both academic standards and LGU regulations.

Fieldwork and Data Collection Notes. Data collection for the study utilized a combination of methods, including route frequency observations, travel-time recordings, and vehicle route inventories. The researchers used the official route data provided by the City Planning and Development Offices of Trece Martires City and General Trias City, which served as the primary reference for the modeling and simulation processes. These official datasets ensured that the representation of the transport network in the MCGA framework accurately reflected the current routes and commuting structures recognized by the local governments. Ethical research protocols, respondent confidentiality, and continuous coordination with the respective LGUs guided all stages of fieldwork, ensuring that data collection and analysis were conducted in accordance with institutional and local government standards.

Research Instrument

The study will employ three sets of instruments to assess the NaviCav system: algorithm testing, user evaluation, and expert review. These instruments will be designed to gather both technical and usability data relevant to the system's performance and user experience.

For the algorithm testing, quantitative metrics will be utilized to evaluate the efficiency and accuracy of the Monte Carlo–Guided Genetic Algorithm (MCGA) implementation.

The evaluation will include measures such as average travel time, number of transfers, computation time, and route optimality ratio. The outcome measures will be derived using validated Local Public Transport Route Plans (LPTRPs) of Trece Martires and General Trias, and will be compared against benchmark algorithms such as A* and the standard Genetic Algorithm (GA) to determine relative performance and optimization quality.

The user evaluation will consist of standardized surveys that will be completed by commuters after utilizing the NaviCav web and mobile applications. The System Usability Scale (SUS) will be used to measure ease of use, efficiency, and user satisfaction, while an evaluation form based on ISO/IEC 25010 will be used to assess the system's functionality, usability, reliability, and performance efficiency. Both surveys will utilize a five-point Likert scale to capture the qualitative perceptions and overall experience of the users.

For the expert review, selected IT professionals and transport specialists will evaluate the system using a System Evaluation Checklist and an Acceptance and Evaluation Form, both of which will be structured around the ISO/IEC 25010 software quality model. The experts will assess the system's functionality, reliability, performance efficiency, and usability, providing both quantitative ratings and qualitative feedback.

Through the combination of these instruments, the study will obtain a comprehensive evaluation of the NaviCav system, ensuring that both its algorithmic accuracy and user-centered design are systematically measured and analyzed.

Research Instrument Validation

Prior to the full implementation of the study, all research instruments will be validated to ensure their usability, reliability, and suitability for evaluating the NaviCav system. The validation process will consist of expert review, pilot testing, and reliability analysis to confirm that the instruments effectively measure both technical and user-centered aspects of the system.

For content validity, three domain experts will review the instruments to ensure that all items are clear, consistent, and aligned with the study objectives. Their feedback will be used to refine question wording, structure, and scale uniformity to enhance clarity and logical flow.

The pilot testing will be conducted with the Information and Communications Technology Department (ICTD) of the General Trias City Hall, which will serve as the proposed client and primary beneficiary of the system. The ICTD staff members will be selected as pilot testers due to their direct involvement with local transport systems and their familiarity with implementing digital solutions for public services. Their participation will help ensure that the instruments are practical, understandable, and applicable to real-world use. Feedback gathered during this phase will be used to revise the survey layout, structure, and phrasing to improve clarity and usability.

For reliability testing, the researchers will utilize Cronbach's Alpha statistical analysis to measure the internal consistency of the System Usability Scale (SUS) and the ISO 25010-based instruments. A reliability coefficient of 0.70 or higher will indicate that the instruments are dependable and suitable for use in the main data collection phase.

In summary, the research instruments will undergo expert validation, pilot testing with the ICTD of General Trias City, and reliability analysis to ensure that they are clear, relevant, and reliable for evaluating both the technical quality and user-centered performance of the NaviCav system.

Data Gathering Procedure

The data gathering procedure for this study will be conducted systematically in Trece Martires City and General Trias City to support the development and future evaluation of the NaviCav system. The process will involve collecting, validating, and analyzing transportation-related data through coordination with local government offices, informal interviews, and integration of data analytics from the system itself. The research team will ensure that all procedures adhere to academic and ethical standards, emphasizing data reliability, accuracy, and representativeness.

During the preparation stage, the research team conducted preliminary coordination and information-gathering activities with key municipal offices. Informal interviews were conducted with the City Planning and Development Offices of Trece Martires City and General Trias City to obtain contextual information about existing transportation routes and planning frameworks. The team also formally requested the Local Public Transport Route Plan (LPTRP) documents to secure official data for route verification and analysis. Furthermore, informal interviews were carried out with the

Traffic Management Offices in both cities to gather insights on route operations and traffic flow management. To complement these, the team consulted with the Information and Communications Technology Departments (ICTD) to gain technical insights related to the potential implementation and integration of the NaviCav system. These preparatory activities ensured that the project's data framework was aligned with local transportation realities and supported by verified route information.

Following the preparatory phase, the research team will utilize the NaviCav system's built-in data analytics to collect and process transportation data in real time. The system will gather route usage data from users' interactions and movement patterns, which will serve as the primary dataset for analysis. Through this approach, the research will employ spatial analytics techniques, particularly heatmap visualization, to identify common passenger loading and unloading zones. These heatmaps will provide visual representations of passenger density and movement frequency, allowing the team to pinpoint high-traffic areas and recurring transport patterns. The data collected through this method will be used to generate actionable insights for optimizing route design and evaluating the efficiency of transport flow across key areas in both cities.

Quality assurance procedures will be applied throughout the data analytics process to ensure the integrity and validity of the collected data. System logs and location-based records will be periodically reviewed to eliminate duplicates, outliers, and erroneous entries. Ethical protocols will also be strictly observed; no personal or identifiable user data will be collected or disclosed. All information will be anonymized and aggregated prior to analysis to ensure compliance with data privacy standards.

The data management process will follow a structured plan for secure storage, backup, and documentation. Data files will be stored in encrypted local drives with restricted access to authorized research personnel. Final datasets and visual outputs, such as heatmaps and route activity summaries, will be exported in standardized formats accompanied by appropriate metadata.

Through this systematic data gathering procedure, the research team will obtain verified, analytics-based transport data that will form the foundation for evaluating and improving the NaviCav multi-modal route planning system. The integration of real-time analytics and spatial visualization will enable a more comprehensive understanding of commuter behavior and route efficiency within Trece Martires City and General Trias City.

Testing Plan and Procedure

The testing plan for the NaviCav system will include functionality, performance, and usability testing, which will be conducted consecutively, to ensure that each system module meets functional and performance requirements for design and operation. Specifically, the testing phase will consist of three interrelated procedures designed to validate the system's overall accuracy, stability, and user acceptance:

1. **Module-Level Testing** - Each component of the system, including the frontend, APIs, the Monte Carlo–Guided Genetic Algorithm (MCGA), and the database, will be tested individually. This phase will ensure that each module functions as expected, accepting valid inputs and producing accurate outputs without errors. The performance testing at the module level will evaluate each component's responsiveness and computational efficiency, particularly focusing on the optimization engine's ability to identify near-optimal commuting routes within its runtime specifications. This process will also allow debugging and incremental improvements to each module before full integration.
2. **Integration Testing** - After the module-level testing, integration testing will be conducted to verify that all system components communicate and operate seamlessly as a single functional unit. This process will focus on end-to-end scenarios in which the user will input travel parameters, the system will compute route options using the MCGA, and the final results will be displayed within a specified response time. Integration testing will also include stress tests under simulated peak-load conditions to evaluate the system's stability, responsiveness, and overall performance during simultaneous routing requests.

3. **User Acceptance Testing (UAT) -** Following the deployment of the working prototype, User Acceptance Testing (UAT) will be conducted with selected commuters and expert evaluators from Cavite. This phase will measure the system's real-world usability, reliability, and overall satisfaction using both subjective and objective criteria. The System Usability Scale (SUS) and expert assessments based on ISO 25010 software quality standards will serve as the primary evaluation instruments. The results of the UAT will be analyzed to identify final refinements before the system's broader deployment.

Evaluation Plan and Procedure

The assessment of the NaviCav system will be performed using the ISO 25010 software quality model driving its evaluation according to the three main criteria: Functionality, Usability, and Efficiency. This approach encompasses both technical performance and user experience to provide an exhaustive evaluation of the system's quality.

Functionality. This criterion evaluates the extent to which the functionality of NaviCav meets the specified requirements and user needs. Evaluators will check for accuracy, completeness, and appropriateness of the system routing plan performance, including a breakdown of the enabled multi-modal transport modes and eco-aware routing cluster option.

Usability. This criterion evaluates how effectively and satisfactorily end-users can interact with NaviCav in their roles as commuters. Commute participants from Cavite will provide standardized responses according to the System Usability Scale (SUS) measuring ease of learning the system, ease of operability, error recovery, and aesthetics of the user interface.

Efficiency. This criterion measures the ability of NaviCav to effectively use computational resources in expected operating conditions. This factor will include the

run-time performance of the Monte Carlo–Guided Genetic Optimization algorithm and the system's responsiveness to peak user loads.

Evaluation Methods

- Technical Assessment: Technical evaluation of NaviCav's functionality, usability, and efficiency will be undertaken systematically via expert assessments, using checklists aligned to the ISO 25010 standard.
- User Assessment: End-user assessments (e.g., SUS and interviews) will be used to gather evidence of practical usability and user experience satisfaction.

Therefore, these assessment components together align with comparable international standards for software quality and demonstrate NaviCav's merit as a robust route planning product with technical quality and user-centered quality.

System Development Process

The Software Development Life Cycle (SDLC) is a structured framework to guide the planning, development, testing, implementation, and maintenance of software systems. The primary goal of the SDLC is to deliver quality software to users through well-defined phases. The researchers applied Agile principles of incremental software development, continuous improvement, and flexibility to adjust to requirements as they arose throughout development.

The Agile principles have provided the framework for the researchers to work in cycle increments, which allows permitting the team to rehearse the software project periodically and modify the project outcomes based on feedback and new requirements. The researchers communicated in real-time daily through Discord, to provide members with updates on progress, obstacles and next plans. Daily communication encouraged collaboration and ensured all members were on the same page for project outcomes and priorities. Version control and file modifications were done with GitHub, as source code and documentation of the software is kept on GitHub. The researchers collaborated together in GitHub branches, and used GitHub pull requests. This is a clear, and responsible way to develop the software and project deliverables.

Figure 2

Agile Model of NaviCav: A Web–Mobile Route Planner for Multi-Modal Commuting in Cavite Using Monte Carlo–Guided Genetic Optimization.

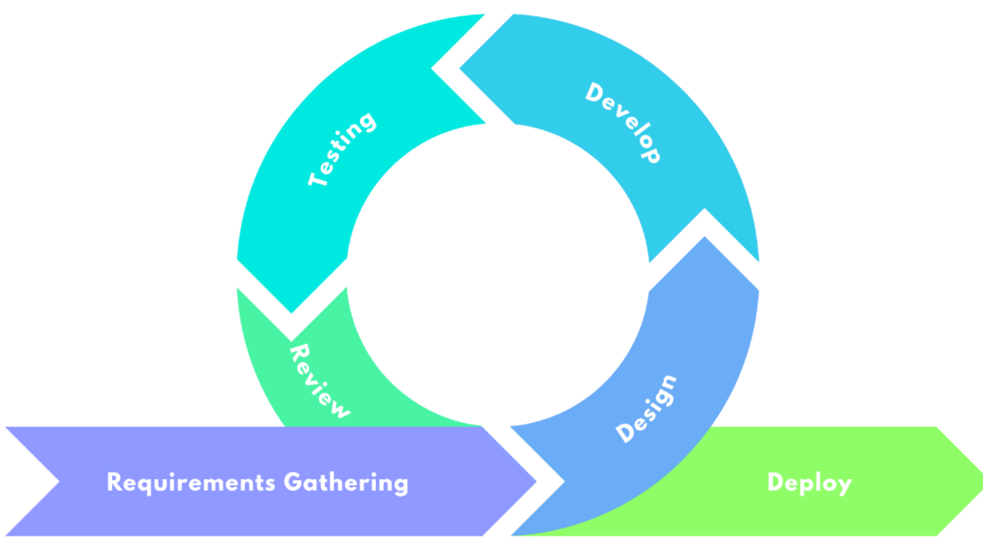


Figure 1 presents the Agile development lifecycle to be applied for NaviCav is iterative and composed of six essential stages: Requirements Gathering, Design, Development, Testing, Review, and Deployment. In the first stage, Requirements

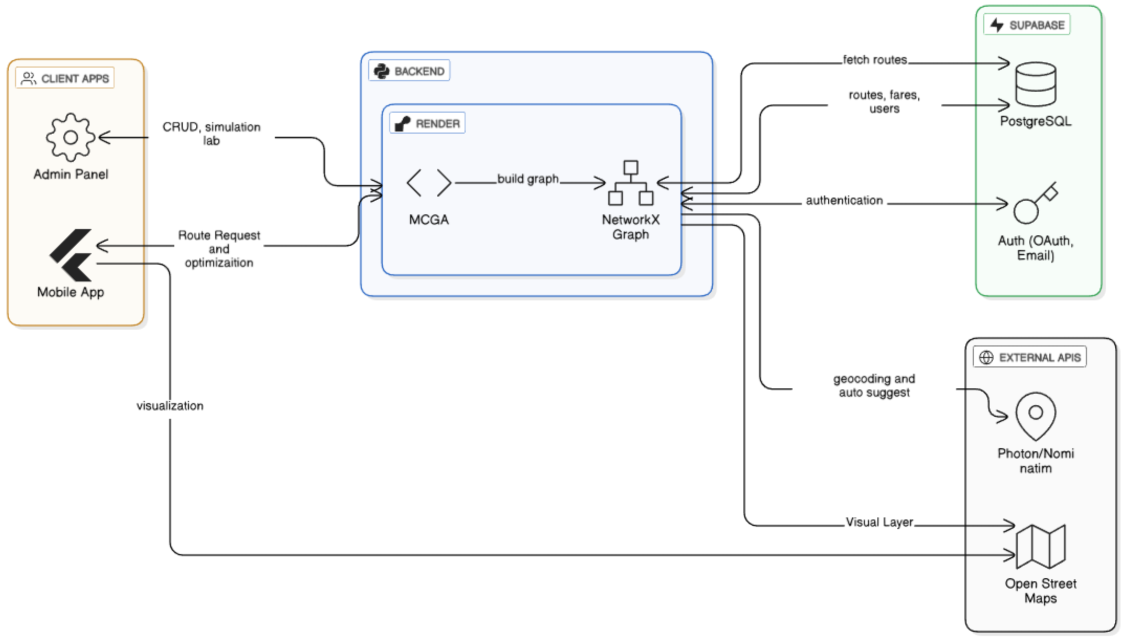
Gathering, the focus will be on data acquisition that includes transport routes, client requirements, and user needs for Cavite's multi-modal commuting area. Next, Design will be concerned with system conceptualization and vision, including the creation of user interface mock-ups, data models, and MCGA workflow specifications. The Development phase will be responsible for the creation of the NaviCav web–mobile platform, the connection of outside services (like geocoding and maps), and the programming of the MCGA optimization engine. The Testing stage will check the

functioning, performance, and accuracy of the route that is being recommended, whereas the Review stage will consider the input of the stakeholders and the pilot users to improve the features and clear up the problems. These phases will go through the process of iteration until the system is ready for Deployment, at which point, the stable version of NaviCav will be released for the real-world and further iterations will be planned as new data and commuter needs arise.

System Architecture

Figure 3

NaviCav's System Architecture



NaviCav's system architecture is designed to support an integrated web–mobile platform for multi-modal commuter route optimization. The architecture uses **Flutter** framework to make the entire system cross-platform, therefore, it deploys the mobile application and the admin management panel together. System interactions are mainly by way of a visual map interface, with **Flutter Maps** using OpenStreetMap Data as the spatial visualization base layer. Besides, geocoding and auto-suggest services are backed up by **Photon** and **Nominatim**, which work on the **OpenStreetMap** platform facilitating address search and location discovery.

Supabase manages data storage and user authentication; it is a NoSQL database that is very flexible and supports the **PostGIS extension** of PostgreSQL very well, thereby leading to a spatial dataset handling that is very efficient. The backend uses

NetworkX, a graph modeling Python library, for the creation and modification of the potential routing networks that are saved in the database. The selection of the candidate paths is carried out by a **Monte Carlo-Guided Genetic Algorithm (MCGA)** that takes into account the criteria that the commuters have chosen, such as the fastest, the cheapest, or the most ecologically sound route, and assigns a rank to each path accordingly.

Figure 4

NaviCav's Context Diagram

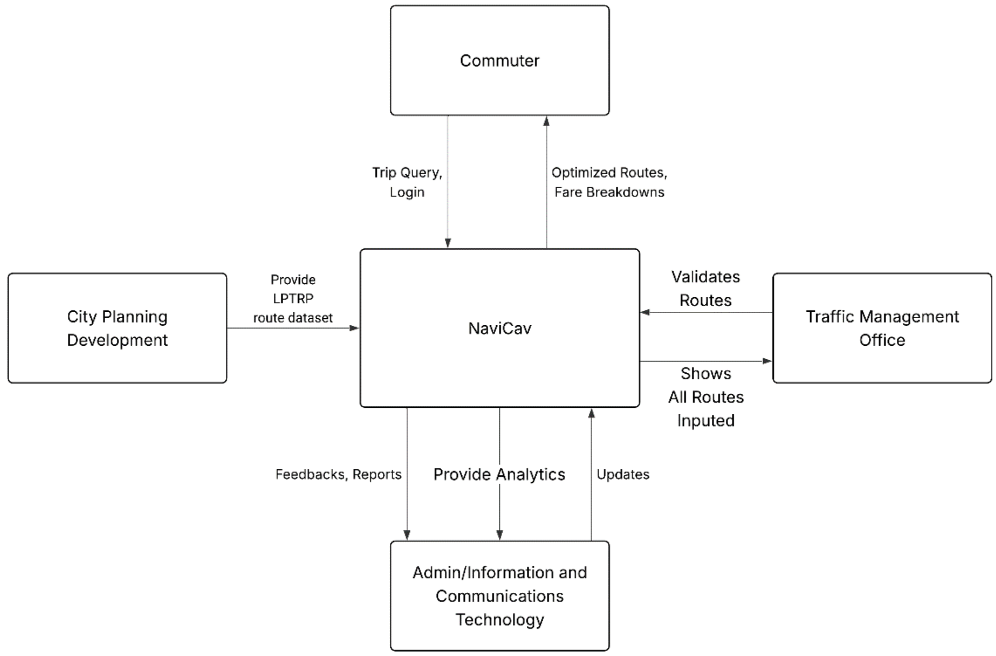
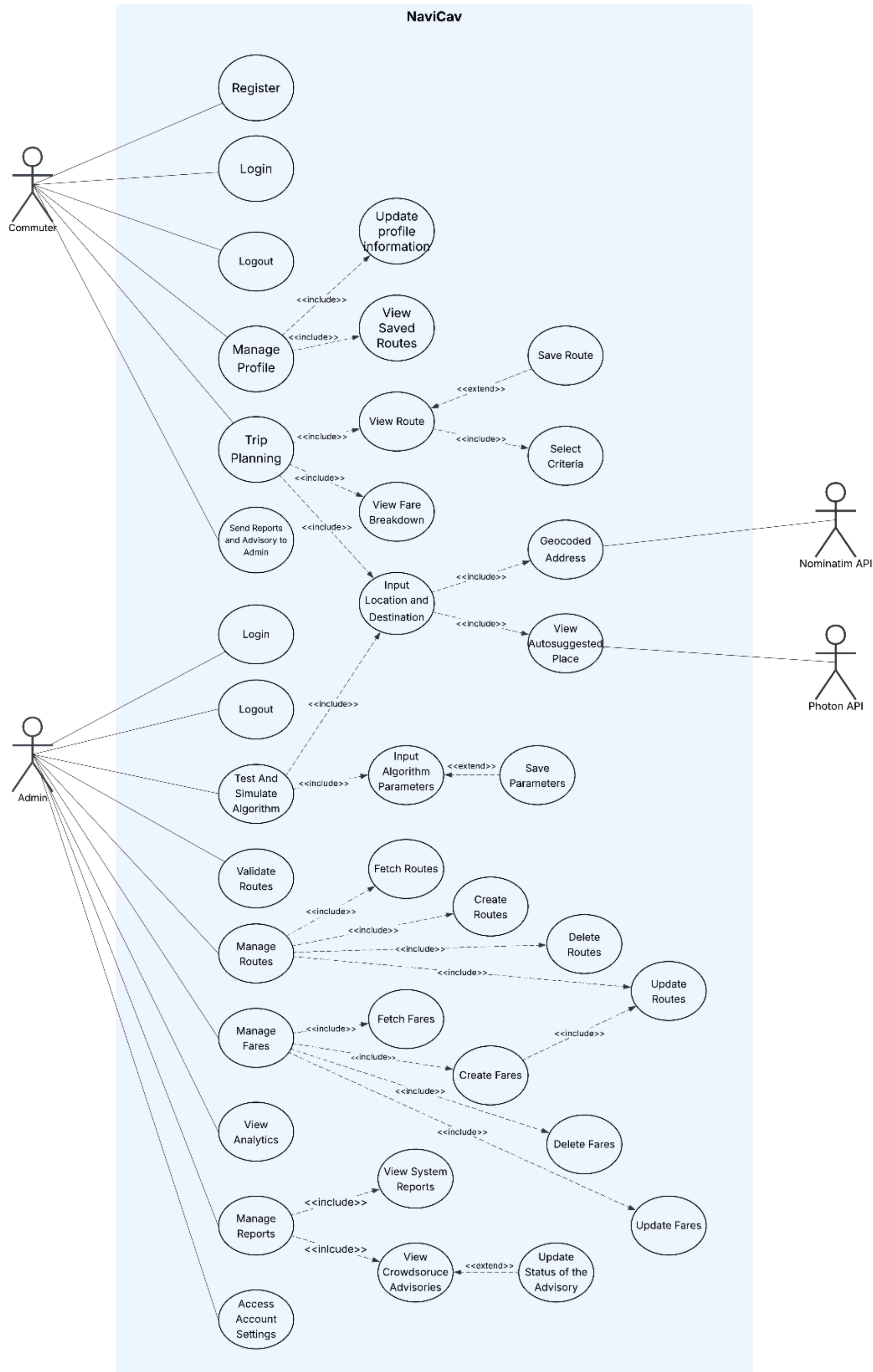

Commuter
Trip Query,
Login
Optimized Routes,
Fare Breakdowns
City Planning
Development
Provide
LPTRP
route dataset
NaviCav
Validates
Routes
Traffic Management
Office
Shows
All Routes
Inputed
Feedbacks, Reports
Provide Analytics
Updates
Admin/Information and
Communications
Technology

Figure 5



NaviCav's Use Case Diagram

Figure 5 presents the Use Case view of the NaviCav platform, showing the interaction between the commuters, the system, the administrators, and external APIs. The commuter actor mainly engages in account-related tasks (registration, logging in, logging out, and managing the profile), trip planning, and exploring different routes. Trip planning involves specifying the starting and ending points, seeing autosuggested places and geocoded locations through the Photon and Nominatim APIs, setting optimization criteria, viewing routes and cost breakdowns, and optionally storing the favorite routes.

The admin actor is involved in managing the system and assuring its quality. The administrators have the power to set up, modify, and close down routes and fares, import the routes created by the MCGA engine, and also manage and test the parameters of the algorithms and the optimization process until changes are done to the commuters. They also get to see the statistics and the reports of the system, take care of the advisories from the crowd, and handle the accounts and data management tasks.

The diagram employs «include» and «extend» relations to show that high-level objectives, such as Trip Planning or Manage Routes, are made up of more detailed sub-functions. This unified view illustrates that NaviCav combines user-oriented trip planning, admin configuration, and external geocoding services into a single route-planning workflow without the need for a separate explanation for each individual use case.

Data Analysis

Data analysis is structured to fulfill the study's objectives of designing, developing, and validating the NaviCav platform and its MCGA optimization engine. The process involves several stages, described below.

1. Data Preprocessing

Data preprocessing will support the integration of multimodal transport datasets from Trece Martires City and General Trias City. Geographic coordinates will be transformed into a common spatial reference system to ensure consistency of spatial operations. Event based data, including boarding and alighting counts, will be aggregated into uniform grid cells to facilitate density analysis. Numerical attributes such as travel time, fare cost, and passenger count will be normalized using min–max scaling to ensure comparability when used in multi criteria fitness evaluations. The normalization will follow the expression

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

where x represents the original value and x' represents the scaled value. This approach is aligned with established transport analytics workflows that support visualization and optimization pipelines (Xu et al., 2024; Jiang et al., 2023).

2. Monte Carlo Guided Genetic Optimization Algorithm Evaluation

Route planning will be framed as a Search Based Software Engineering problem in which Monte Carlo sampling generates diverse initial candidate routes that the genetic algorithm will subsequently evolve. Each candidate route will be assessed through a multi criteria fitness function incorporating travel time T, monetary cost C, and estimated carbon emissions E, using the formulation

$$F = w_t(\frac{1}{T}) + w_c(\frac{1}{C}) + w_e(\frac{1}{E})$$

where w_t, w_c, and w_e are nonnegative weights reflecting commuter or policy priorities. Monte Carlo sampling will introduce stochastic variation that reflects uncertainties in informal public transport systems, including traffic conditions and vehicle arrival patterns. The genetic component will refine these candidates through selection, crossover, and mutation, consistent with established evidence that hybrid Monte Carlo evolutionary strategies improve exploration and exploitation balance (Hebbar, 2023; Castro Amoedo et al., 2024).

To reflect real world variability, each candidate route will undergo multiple Monte Carlo simulations, typically around thirty repetitions. During each simulation, travel and waiting times will be sampled from empirical or estimated distributions that reflect uncertainties such as congestion and unpredictable vehicle headways. This repetition will produce route reliability metrics including mean travel time, variance, and confidence intervals. These outputs will allow the study to evaluate the stability of routes under hypothetical disturbances such as temporary closures or sudden congestion. This stochastic evaluation will ensure that NaviCav recommends not only the shortest routes but also the most reliable ones given the dynamic nature of the transport network.

The analysis will also incorporate parameter tuning procedures for the MCGA. Population sizes between fifteen and thirty individuals will be evaluated, along with mutation rates between 0.2 and 0.4 and crossover rates between 0.65 and 0.85. Monte Carlo sample sizes between twenty and fifty simulations will be examined to identify a balance between computational feasibility and stability of fitness estimates. Convergence criteria will include stopping conditions such as maximum generation counts and fitness plateau detection, where convergence will be inferred if the relative improvement in fitness falls below approximately one percent over several consecutive generations. These procedures are drawn from the attached parameter tuning and convergence criteria document and from evolutionary computation literature.

The comparative performance of MCGA will be analyzed with respect to three baseline algorithms: A*, the standalone Genetic Algorithm, and Particle Swarm Optimization. A* will serve as a deterministic benchmark that performs efficiently on static networks but lacks the capability to model stochastic variations. The standalone Genetic Algorithm will serve as a non-hybrid comparator that demonstrates the effect of removing Monte Carlo sampling. Particle Swarm Optimization will represent a heuristic approach strong in global search but comparatively weaker in local refinement. MCGA is expected to outperform these baselines according to metrics such as travel time, travel time variability, reliability under uncertainty, and robustness across repeated trials, based on empirical patterns observed in the pathfinding comparative analysis dataset.

3. Spatial Analytics

Spatial analytics will be performed to visualize boarding and alighting patterns, evaluate demand density, and validate routing outputs. For each grid cell i, the normalized passenger density D_i will be computed using

$$D_i = \frac{n_i}{n_{max}}$$

where n_i represents the event count in the grid cell and n_i represents the maximum observed event count. These density maps will help identify high demand corridors, inform weighting strategies within the fitness function, and validate the spatial coverage of recommended routes. Contemporary methods for large scale heatmap generation and flow estimation will guide these analyses (Xu et al., 2024; Jiang et al., 2023).

4. Computational Performance and Scalability

The optimization engine will be evaluated through controlled experiments that measure mean runtime, runtime variability across repeated trials, generations required to achieve convergence, best fitness achieved per unit of wall clock time, and peak memory usage. Scalability analyses will evaluate the behavior of the optimization engine as the number of transport stops, potential routes, or simulated users increases. Adaptability will be assessed by introducing dynamic perturbations such as unexpected delays, route closures, or surges in passenger demand to evaluate reoptimization speed and solution degradation. These procedures follow accepted evaluation practices for optimization algorithms and Monte Carlo simulation frameworks (Castro Amoedo et al., 2024; Hebbar, 2023).

5. Software Quality Evaluation

Software quality will be evaluated using the ISO/IEC 25010:2023 quality model. The evaluation will cover functionality, reliability, usability, performance efficiency, and maintainability. Quantitative metrics will include automated test pass rates, mean time between failures under simulated loads, user task success rates, task completion times, and throughput during stress testing. Qualitative evaluations will emerge from expert reviews and thematic analysis of user feedback. The use of ISO/IEC 25010 will ensure alignment with current international standards (ISO/IEC, 2023; Canlas et al., 2021).

6. Statistical Analysis

Statistical analysis will be applied to examine relationships between observed demand patterns and route attributes and to assess the comparative performance of algorithmic variants. Depending on data characteristics, correlations will be measured using Pearson’s correlation or appropriate nonparametric alternatives. Hypothesis testing will be conducted using assumption checks, corrections for multiple comparisons, and effect size calculations. Cross validation and repeated Monte Carlo experiments will be employed to support robustness and generalizability, in line with established SBSE and evolutionary computation methodologies (Katoch et al., 2021; SSBSE proceedings, 2021).

7. Sensitivity and Scenario Analysis

To rigorously assess the robustness and applicability of the Monte Carlo–guided Genetic Algorithm (MCGA) in informal transport contexts, the analysis will proceed along two complementary axes: scenario testing under varying operational conditions and sensitivity testing of core algorithmic parameters.

First, scenario testing will simulate realistic demand and network-disruption conditions to evaluate algorithm adaptability. Three demand scenarios will be considered: (1) peak hour demand, in which passenger volumes increase by 150 percent and service frequencies decline by 120 percent; (2) off peak demand, modelling sparse service with headways magnified by 150 percent; and (3) network disruption, wherein 10 percent, 25 percent, and 40 percent of routes are randomly disabled. For each scenario, the MCGA will be run multiple times (e.g., 10 independent runs per origin–destination pair, across a representative set of O–D pairs in urban, suburban, and periphery zones) to measure key performance metrics such as travel time change ratios, transfer counts, route feasibility, and recommendation consistency. This enables the quantification of how service conditions (demand surge or route closure) impact route quality, stability, and connectivity.

Second, sensitivity testing will explore how variations in GA control parameters influence convergence behavior, solution quality, and computational cost. The analysis will include: (a) population size, tested across a range (e.g., 10–50 individuals) while holding crossover, mutation, and Monte Carlo (MC) sampling constant; (b) mutation rate, varied systematically (e.g., 0.10–0.50) to examine trade-offs between exploration and premature convergence; and (c) Monte Carlo sample size, exploring sample sizes from 10 to 100 simulations to assess how estimation noise, ranking stability, and computational scaling change with sample size. For each configuration, the MCGA will be executed in multiple independent replicates (e.g., 20 runs) to compute metrics such as generations to convergence, fitness mean and variance, runtime, memory usage, and ranking consistency.

Moreover, robustness validation will explicitly address uncertainties unique to informal transport systems. Vehicle-availability simulations will model operator absence probabilistically (e.g., 5–30 percent unavailability) and evaluate how often recommended routes require fallback options, as well as how solution quality deteriorates under such unpredictability. Similarly, route modification tolerance will be tested by injecting random perturbations into network geometry (e.g., coordinate offsets) and re running MCGA to measure top k recommendation stability and degradation in travel metrics. Finally, service-interruption recovery will be examined by inducing simulated service failures (e.g., route shutdowns) and timing the re-optimization process, with attention to re-optimization latency and relative solution quality.

From the sensitivity experiments, informed by the scenario testing, a set of conservative parameter configurations will be identified that balance solution quality, diversity, and computational practicality. For example, a moderate population (≈ 20–25), mutation rate in the 0.25–0.35 range, crossover rate around 0.70–0.80, and MC sample sizes tailored to deployment context (e.g., 20–30 for real time use, 40–50 for planning, 60–70 for offline optimization) will be recommended, consistent with the expected trade off between robustness and optimization efficiency.

Statistical rigor will be maintained by reporting all key metrics with 95 percent confidence intervals, applying multiple-comparison correction (for example Bonferroni adjustment), and reporting effect sizes. Random seeds for both evolutionary and Monte Carlo processes will be systematically documented and version-controlled, ensuring full reproducibility of experiments.

This combined scenario–sensitivity analysis framework will demonstrate that MCGA is not only effective under nominal conditions but remains robust under stressors typical of informal transport systems, notably demand fluctuations, service disruptions, and stochastic operator behaviour. It will also yield parameter recommendations aligned to different deployment modalities (mobile, planning, offline), thereby supporting practical implementation by planners and practitioners.

8. Reproducibility and Privacy

Reproducibility will be ensured by maintaining all preprocessing scripts, optimization configurations, algorithm parameters, and visualization code in a version-controlled repository. Random seeds for Monte Carlo sampling and genetic

operations will be explicitly stored to ensure repeatable stochastic behavior. Detailed metadata will accompany all generated outputs, including travel time simulations, fitness evaluations, and heatmaps. Privacy will be maintained by aggregating spatial outputs and anonymizing sensitive trip data, with access restrictions applied to raw mobility logs. These practices will ensure transparency, repeatability, and verifiability, enabling future researchers to replicate or extend the methodology with confidence.

To maintain the reproducibility of the results, all the code, scripts, and datasets used in optimization experiments are stored on GitHub under version control. This encompasses data cleansing, fitness function evaluations, Monte Carlo–Genetic Algorithm as well as the related configuration files. The random seeds are assigned explicitly for Monte Carlo sampling and genetic algorithm operations to ensure the same and repeatable stochastic behavior, thus making the verification and debugging easier. Detailed metadata is kept for all outputs such as route plans, travel time simulations, fitness scores, and heatmaps. This metadata specifies input parameters, algorithm versions, and environmental conditions for complete traceability. Such rigorous practices greatly enhance the openness, repeatability, and verifiability of the research thus future researchers would not only be able to replicate the experiments but also to validate the findings or extend the methodology with confidence.

Ethical Considerations

In a CS research, ethical considerations play a significant role in ensuring that the project is conducted responsibly and aligns with professional and societal standards. Here are key ethical considerations to keep in mind:

1. Data Privacy and Confidentiality

- **Sensitive Information Protection**: All user and location data collected by NaviCav will be handled with strict confidentiality to safeguard personal information.

- **Data Anonymization**: Location traces, GPS data, and usage logs will be anonymized and aggregated to prevent the identification of individual users.

- **Compliance with Data Protection Laws**: Adherence to laws such as GDPR (General Data Protection Regulation) or HIPAA (Health Insurance Portability and Accountability Act) is essential when handling sensitive data.

2. Informed Consent

- **Participants' Knowledge**: Any human subjects involved in surveys, usability testing, or pilot deployments will be provided clear information about the study's purpose, data collected, and their rights.

- **Voluntary Participation**: Participation will be strictly voluntary, with participants free to withdraw at any point without consequence.

3. Intellectual Property and Plagiarism

- **Original Work**: Research projects should avoid plagiarism, ensuring proper citation of sources and acknowledgment of previous research.

- **Copyright and Licensing**: Any Use of third-party software, libraries, or tools should respect copyright and licensing agreements, including proper credits within the application if needed.

4. Cybersecurity and Protection from Harm

- **Ensuring Security**: NaviCav's software will be developed with security best practices to protect users from cyber threats and unauthorized data access.

- **Avoiding Harm**: The project commits to avoiding negligence or vulnerabilities that could harm users or systems.

5. Bias and Fairness

- **Algorithmic Bias**: The MCGA and related algorithms will be evaluated to ensure they do not systematically disadvantage certain routes, user groups, or regions.

- **Transparency**: The methodology and data usage will be clearly documented to enable scrutiny and prevent hidden biases.

6. Accountability and Integrity

- **Accuracy in Reporting**: Findings, including limitations and potential weaknesses in data or algorithms, will be reported transparently and accurately.

- **Responsibility for Actions**: The research team takes responsibility for the ethical conduct of the study throughout all phases.

7. Collaboration Ethics

- **Equal Credit**: All collaborators from the multiple city government departments and any other contributors will receive appropriate credit to ensure that everyone's contributions are properly acknowledged.

- **Conflict of Interest**: Any potential conflicts of interest, whether personal, financial, or professional, that could affect the research outcomes or interpretations will be disclosed transparently.

These ethical considerations ensure that CS research is conducted responsibly, safeguarding the interests of individuals, society, and the broader tech community.

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V. N., Department of Information Technology, St. Paul University Philippines, Tuguegarao City (Cagayan), Philippines., Gumarang, Dr. S. G., & Department of Information Technology, St. Paul University Philippines, Tuguegarao City (Cagayan), Philippines. (2025). TC-Trissea (Tuguegarao City Tricycle Riders Information, Security and Satisfaction Enhancement App): A Proposed Mobile App for Tricycle Commuters of Tuguegarao City. International Journal of Innovative Technology and Exploring Engineering, 14(6), 1–14. https://doi.org/10.35940/ijitee.D1076.14060525

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APPENDICES

APPENDIX A

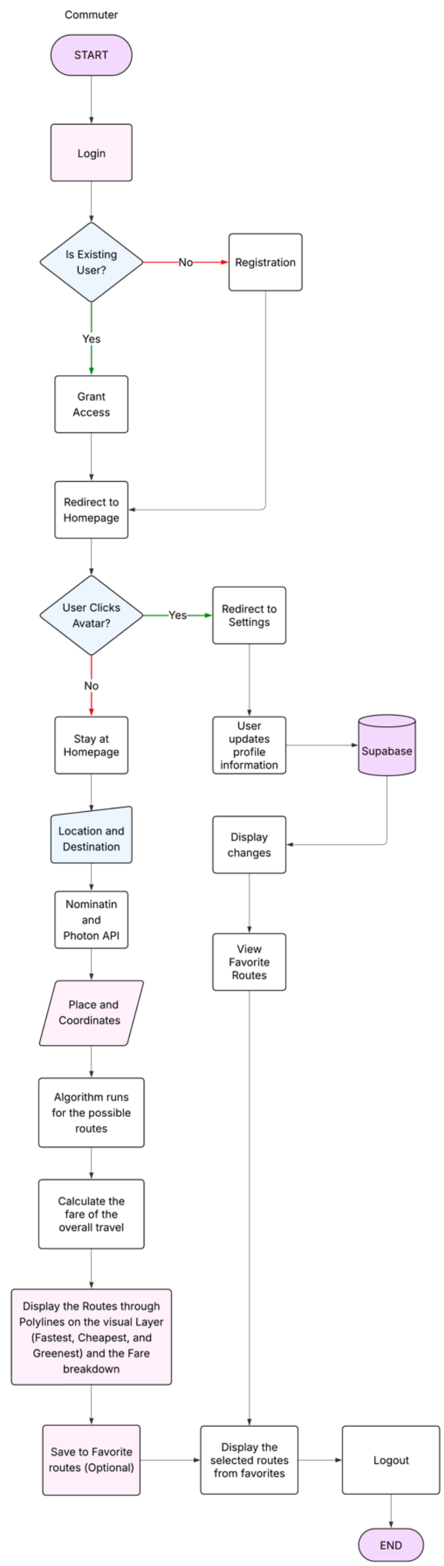
Gantt Chart

Activities	Months

	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	April	May	Jun	July
Planning and Requirement Gathering												
System Design and Architecture												
Backend and Frontend Development												
Training of the Algorithm												
Integration of the Algorithm and the system												
Real-Time Testing and Algorithm Optimization												
Integration and Feature Enhancement												
Testing and Final Optimizations												
Final Revisions and Polishing												
Documentation												

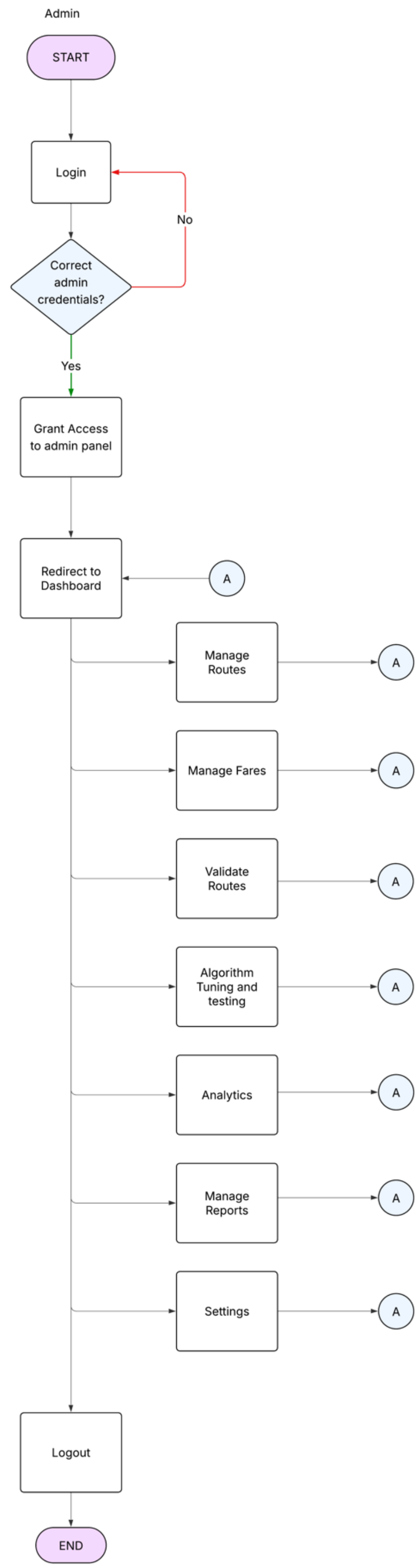
APPENDIX B

User/Commuter Flowchart


Commuter
START
Login
Is Existing User?
No
Registration
Yes
Grant Access
Redirect to Homepage
User Clicks Avatar?
Yes
Redirect to Settings
No
Stay at Homepage
User updates profile information
Supabase
Location and Destination
Display changes
Nominatin and Photon API
View Favorite Routes
Place and Coordinates
Algorithm runs for the possible routes
Calculate the fare of the overall travel
Display the Routes through Polylines on the visual Layer (Fastest, Cheapest, and Greenest) and the Fare breakdown
Save to Favorite routes (Optional)
Display the selected routes from favorites
Logout
END

APPENDIX C

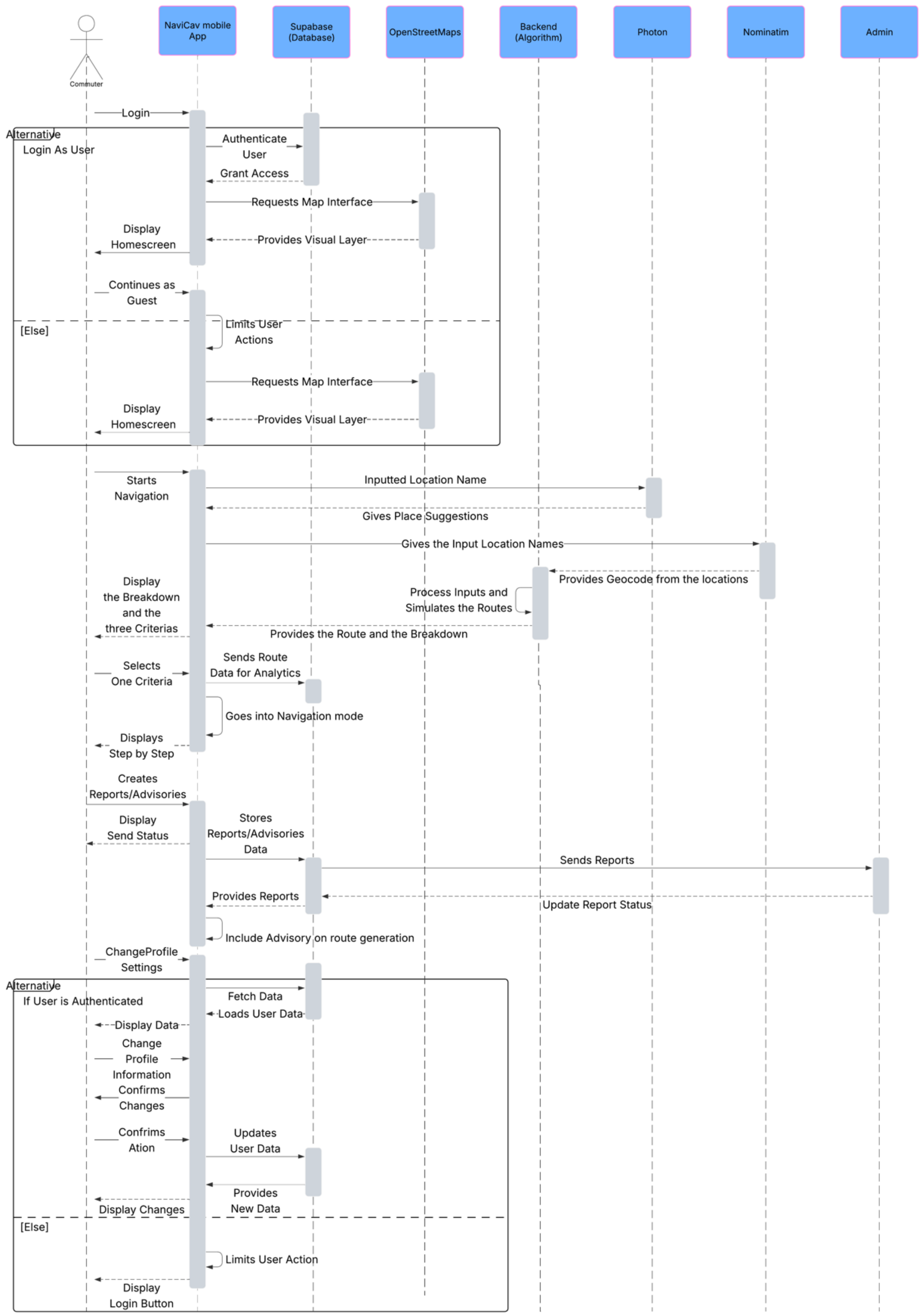
Admin Flowchart



Admin
START
Login
No
Correct admin credentials?
Yes
Grant Access to admin panel
Redirect to Dashboard
A
Manage Routes
A
Manage Fares
A
Validate Routes
A
Algorithm Tuning and testing
A
Analytics
A
Manage Reports
A
Settings
A
Logout
END

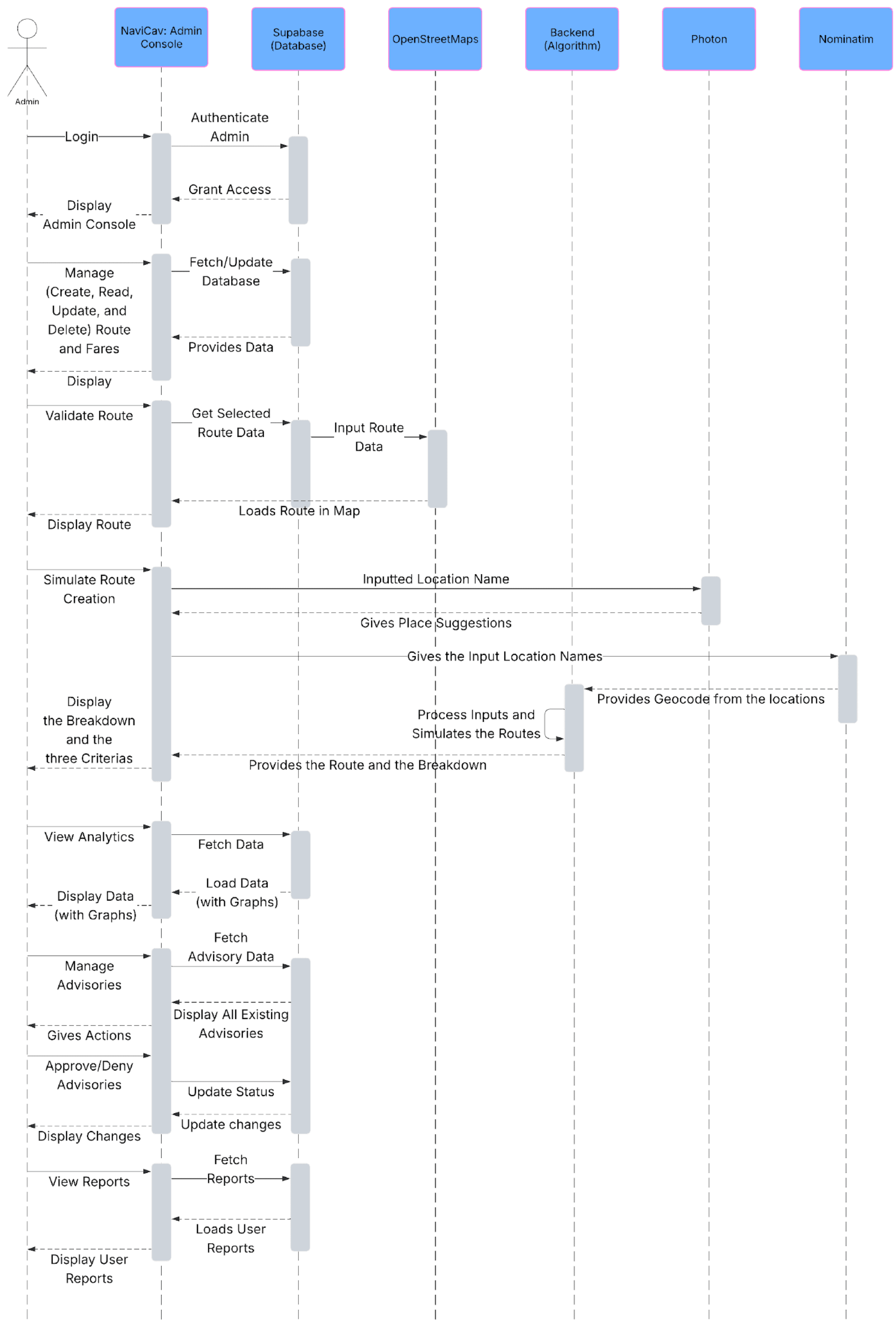
APPENDIX D

User/Commuter Sequence Diagram


Commuter
NaviCav mobile App
Supabase (Database)
OpenStreetMaps
Backend (Algorithm)
Photon
Nominatim
Admin
Login
Alternative
Login As User
Authenticate User
Grant Access
Requests Map Interface
Provides Visual Layer
Display Homescreen
Continues as Guest
[Else]
Limits User Actions
Requests Map Interface
Provides Visual Layer
Display Homescreen
Starts Navigation
Inputted Location Name
Gives Place Suggestions
Gives the Input Location Names
Provides Geocode from the locations
Process Inputs and Simulates the Routes
Display the Breakdown and the three Criterias
Provides the Route and the Breakdown
Selects One Criteria
Sends Route Data for Analytics
Goes into Navigation mode
Displays Step by Step
Creates Reports/Advisories
Display Send Status
Stores Reports/Advisories Data
Sends Reports
Provides Reports
Update Report Status
Include Advisory on route generation
ChangeProfile Settings
Alternative
If User is Authenticated
Fetch Data
Loads User Data
Display Data
Change Profile Information
Confirms Changes
Confrims Ation
Updates User Data
Provides New Data
Display Changes
[Else]
Limits User Action
Display Login Button

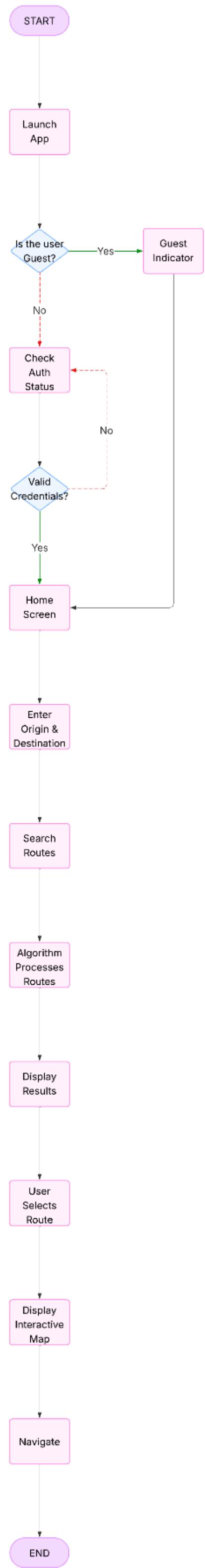
APPENDIX E

Admin Sequence Diagram


Admin
NaviCav: Admin Console
Supabase (Database)
OpenStreetMaps
Backend (Algorithm)
Photon
Nominatim
Login
Authenticate Admin
Grant Access
Display Admin Console
Manage (Create, Read, Update, and Delete) Route and Fares
Fetch/Update Database
Provides Data
Display
Validate Route
Get Selected Route Data
Input Route Data
Loads Route in Map
Display Route
Simulate Route Creation
Inputted Location Name
Gives Place Suggestions
Gives the Input Location Names
Provides Geocode from the locations
Process Inputs and Simulates the Routes
Provides the Route and the Breakdown
Display the Breakdown and the three Criterias
View Analytics
Fetch Data
Load Data (with Graphs)
Display Data (with Graphs)
Manage Advisories
Fetch Advisory Data
Display All Existing Advisories
Gives Actions
Approve/Deny Advisories
Update Status
Update changes
Display Changes
View Reports
Fetch Reports
Loads User Reports
Display User Reports

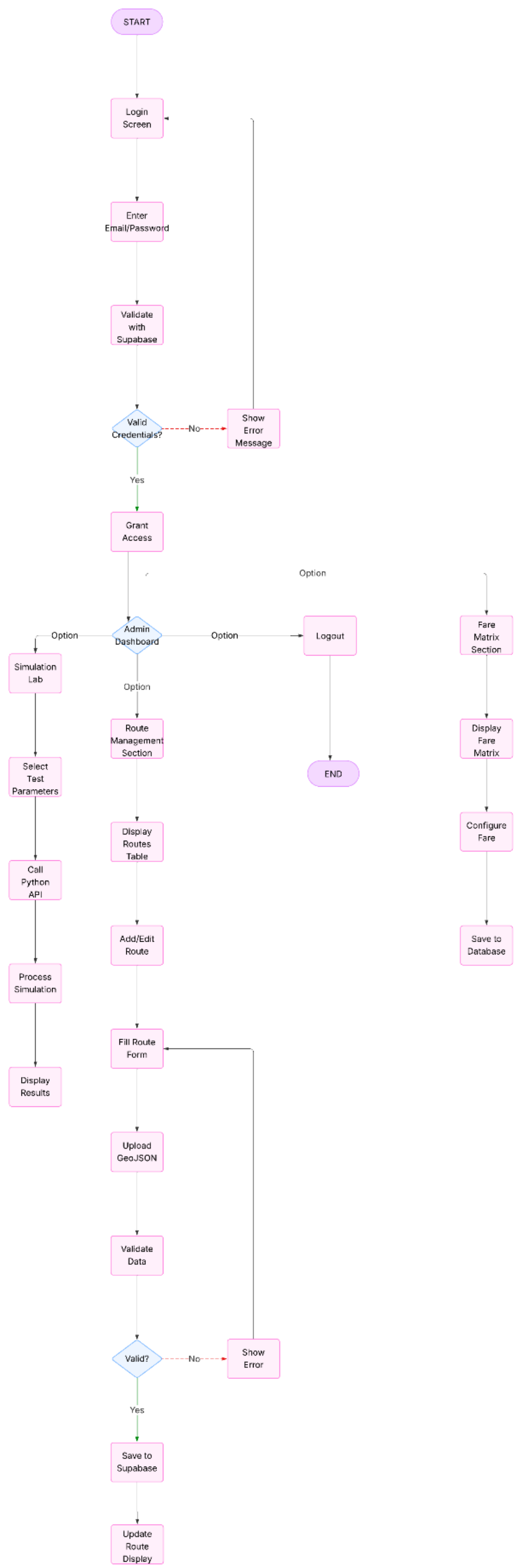
APPENDIX F

User/Commuter Process Flow Diagram


START
Launch App
Is the user Guest?
Yes
Guest Indicator
No
Check Auth Status
No
Valid Credentials?
Yes
Home Screen
Enter Origin & Destination
Search Routes
Algorithm Processes Routes
Display Results
User Selects Route
Display Interactive Map
Navigate
END

APPENDIX G

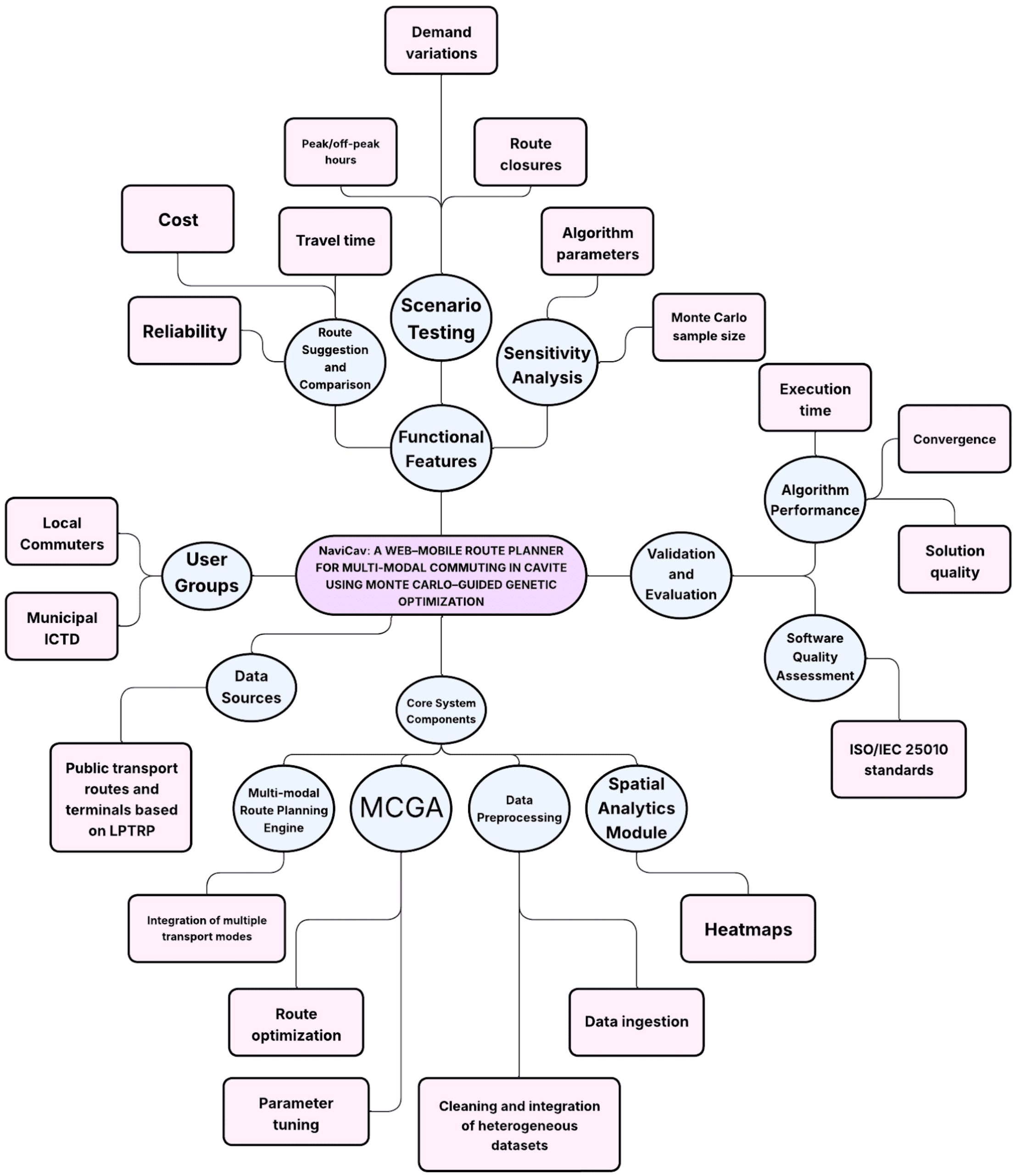
Admin Process Flow Diagram



START
Login Screen
Enter Email/Password
Validate with Supabase
Valid Credentials?
No
Show Error Message
Yes
Grant Access
Admin Dashboard
Option
Simulation Lab
Select Test Parameters
Call Python API
Process Simulation
Display Results
Route Management Section
Display Routes Table
Add/Edit Route
Fill Route Form
Upload GeoJSON
Validate Data
Valid?
Show Error
Save to Supabase
Update Route Display
Logout
END
Fare Matrix Section
Display Fare Matrix
Configure Fare
Save to Database

APPENDIX H

Conceptual Diagram Level 1



NaviCav: A WEB–MOBILE ROUTE PLANNER FOR MULTI-MODAL COMMUTING IN CAVITE USING MONTE CARLO–GUIDED GENETIC OPTIMIZATION
Functional Features
Scenario Testing
Demand variations
Peak/off-peak hours
Route closures
Route Suggestion and Comparison
Cost
Travel time
Reliability
Sensitivity Analysis
Algorithm parameters
Monte Carlo sample size
User Groups
Local Commuters
Municipal ICTD
Validation and Evaluation
Algorithm Performance
Execution time
Convergence
Solution quality
Software Quality Assessment
ISO/IEC 25010 standards
Data Sources
Public transport routes and terminals based on LPTRP
Core System Components
Multi-modal Route Planning Engine
Integration of multiple transport modes
MCGA
Route optimization
Parameter tuning
Data Preprocessing
Data ingestion
Cleaning and integration of heterogeneous datasets
Spatial Analytics Module
Heatmaps

APPENDIX I

General Trias Request Letter for Local Public Transport Route Plan

Sept 09, 2025

Engr. EnP Jemie P. Cubillo
City Planning and Development Coordinator
City Planning and Development Office – General Trias
Provincial Government of Cavite
1st Floor, City Hall General Trias City, Cavite

Dear Engr. Cubillo,

Greetings in *Veritas et Fortitudo Pro Deo et Patria.*

We the Bachelor of Science in Computer Science students are currently conducting a research project entitled "NaviCav: A Web–Mobile Route Planner for Multi-Modal Commuting in Cavite Using Monte Carlo–Guided Genetic Optimization". This requirement is a Thesis 1 course being handled by the undersigned professor.

Specifically, we would like to seek your guidance and possible assistance in understanding the transportation situation within the City of General Trias. If available, we would be grateful for any information or references regarding public utility routes, terminals, stops, or traffic management practices that could support the academic focus of our study.

Rest assured that all information gathered shall be treated with strict confidentiality and for academic purposes only.

Thank you very much and looking forward to your approval regarding this matter.

Sincerely yours,

Lance Christian F. Romero
Project Leader

Noted:

Alyssa Paola Pocaan
Thesis 1 Professor

Endorsed by:

Jerian R. Peren, LPT, PhD
Program Chair, Computer Science

OFFICE OF THE CITY PLANNING AND DEVELOPMENT COORDINATOR
CITY OF GENERAL TRIAS CAVITE
RECEIVED
09 SEP 2025
RECEIVED BY: ENGR. RAYMARK R. RAPOSON
Office of the CPDC

We, the undersigned, received the documents requested from the Office of the City Planning and Development Coordinator – City of General Trias. Upon the receipt of said documents, we would also commit with the following provisions:

- **To use the requested data / files ONLY for the purpose stated in the letter request given to the Office of the City Planning and Development Coordinator**
- **To not share the requested data / files to other users**
- **To credit / use proper citation when referencing or using the requested data**

Signed this 9th day of September, 20 25, at the Office of the City Planning and Development Coordinator, 1st Floor, General Trias City Hall, City of General Trias, Cavite.

Clients / Students:

CPDC Staff: ENGR. RAYMARK R. RAPOSON
Office of the CPDC

APPENDIX J

Letter of Intent General Trias City Government

LPU
LYCEUM OF THE PHILIPPINES UNIVERSITY
MANILA · MAKATI · BATANGAS · LAGUNA · CAVITE · DAVAO

October 13, 2025

Hon. Luis A. Ferrer IV
City Mayor
City Hall General Trias, Cavite

Dear Hon. Ferrer IV,

Greetings in *Veritas et Fortitudo Pro Deo et Patria.*

We are undergraduate students from the College of Information Technology and Computer Science of Lyceum of the Philippines University–Cavite, currently conducting our thesis entitled *"NaviCav: A Web–Mobile Route Planner for Multi-Modal Commuting in Cavite."*

NaviCav is a route planning system designed to help commuters access reliable and convenient travel options while providing local governments with data that may assist in improving transport planning, traffic management, and mobility decisions.

In line with this, we respectfully request your approval and support to establish the City Government of General Trias as our client for this study. With your permission, we hope to coordinate with the **City Planning and Development Office (CPDO)**, **Traffic Management Office (TMO)**, and **Information and Communications Technology Department (ICTD)** for guidance and **conducting interviews for data gathering** related to the city's transport planning, traffic operations, and technology initiatives.

All information gathered will be treated with full confidentiality and used solely for academic purposes. The completed system will also be turned over to the City Government through the ICTD for future reference or potential use.

Attached herewith is our project proposal for your review. Thank you very much for your kind consideration and continued support.

008695

Sincerely yours,

Lance Christian F. Romero
Project Leader

OFFICE OF THE MAYOR
RECEIVED
13 OCT 2025
BY:
CITY OF GENERAL TRIAS, CAVITE
TEL. NO.: (046) 509-1250

Noted:

Alyssa Paola Pocaan. MIT
Thesis 1 Professor

Jerian R. Peren, LPT, PhD
Program Chair, Computer Science

STARS RATING SYSTEM

CAVITE
Governor's Drive, Brgy. Manggahan, City of General Trias,
Cavite, 4107 Philippines
Tel. No.: +6346 481-1400
www.lpu.edu.ph

SGS

APPENDIX K

Letter of Intent for Trece Martires City Planning and Development Office

LPU
LYCEUM OF THE PHILIPPINES UNIVERSITY
MANILA • MAKATI • BATANGAS • LAGUNA • CAVITE • DAVAO

OFFICE OF THE CITY PLANNING AND DEVELOPMENT COORDINATOR
CITY GOVERNMENT OF TRECE MARTIRES
RECEIVED
BY: MJ
DATE: 10/21/2025 TIME: 11:25 am

October 14, 2025

Mr. Alberto S. Ararao
City Planning and Development Coordinator
2nd Floor City Hall Bldg., Brgy. San Agustin, Trece Martires City, Cavite

Dear Sir Ararao,

Greetings in *Veritas et Fortitudo Pro Deo et Patria.*

We are undergraduate students from the College of Information Technology and Computer Science of Lyceum of the Philippines University–Cavite, currently conducting our thesis entitled *"NaviCav: A Web–Mobile Route Planner for Multi-Modal Commuting in Cavite."*

NaviCav is a route planning system designed to help commuters access reliable and convenient travel options while providing local governments with data that may assist in improving transport planning, traffic management, and mobility decisions.

In connection with this, we respectfully seek your office's support and participation as our client for this study. We hope to coordinate with your office for **guidance and data gathering, particularly through an interview,** to better understand the city's **Local Public Transport Route Plan (LPTRP)** and related transport planning initiatives.

All information gathered will be treated with full confidentiality and used solely for academic purposes.

Thank you very much for your time and kind consideration.

Sincerely yours,

Lance Christian F. Romero
Project Leader

Noted:

Alyssa Paola Pocaan, MIT
Thesis 1 Professor

Jerian R. Peren, LPT, PhD
Program Chair, Computer Science

QS STARS RATING SYSTEM

CAVITE
Governor's Drive, Brgy. Manggahan, City of General Trias,
Cavite, 4107 Philippines
Tel. No.: +6346 481-1400
www.lpu.edu.ph

SGS

APPENDIX L

Letter of Intent for Trece Martires Information and Communication Technology Department

LPU
LYCEUM OF THE PHILIPPINES UNIVERSITY
MANILA • MAKATI • BATANGAS • LAGUNA • CAVITE • DAVAO

October 14, 2025

Mr. Glenn C. Agetano
Head, Information and Communications Technology Department
2nd Floor City Hall Bldg., Brgy. San Agustin, Trece Martires City, Cavite

Dear Sir Agetano,

Greetings in *Veritas et Fortitudo Pro Deo et Patria.*

We are undergraduate students from the College of Information Technology and Computer Science of Lyceum of the Philippines University–Cavite, currently conducting our thesis entitled *"NaviCav: A Web–Mobile Route Planner for Multi-Modal Commuting in Cavite."*

NaviCav is a route planning system designed to assist commuters and support local governments through technology-driven solutions for transport planning and mobility management.

We respectfully seek your office's support and participation as our client for this study. In particular, we hope to coordinate with your office for **technical guidance and data gathering through an interview**, as well as to align our system development with the city's existing ICT initiatives. Upon completion, the system will be **turned over to the City Government through your office** for future reference or possible utilization.

All gathered data will be treated with full confidentiality and used solely for academic purposes.

Thank you very much for your time and kind consideration.

Sincerely yours,

Lance Christian F. Romero
Project Leader

Noted:

Alyssa Paola Pocaan, MIT
Thesis 1 Professor

Jerian R. Peren, LPT, PhD
Program Chair, Computer Science

Information and Communication Technology Division
Trece Martires City
RECEIVED
21 OCT 2025
By: ___

CAVITE
Governor's Drive, Brgy. Manggahan, City of General Trias,
Cavite, 4107 Philippines
Tel. No.: +6346 481-1400
www.lpu.edu.ph

QS STARS
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APPENDIX M

Letter of Intent for Trece Martires Public Order & Safety Office

LPU
LYCEUM OF THE PHILIPPINES UNIVERSITY
MANILA • MAKATI • BATANGAS • LAGUNA • CAVITE • DAVAO

October 14, 2025

Mr. Billy Anciro
Head, Public Order & Safety Office
2nd Floor, Command Center Bldg., Governor's Drive, Brgy. Conchu, Trece Martires City, Cavite

Dear Sir Anciro,

Greetings in *Veritas et Fortitudo Pro Deo et Patria.*

We are undergraduate students from the College of Information Technology and Computer Science of Lyceum of the Philippines University–Cavite, currently conducting our thesis entitled *"NaviCav: A Web–Mobile Route Planner for Multi-Modal Commuting in Cavite."*

NaviCav is a route planning system designed to assist commuters in finding efficient routes while also providing local governments with data that can support traffic and mobility planning.

We respectfully seek your office's support and participation as our client for this study. With your permission, we would like to **conduct an interview for data gathering** focused on the city's **traffic operations, route management, and mobility challenges**, which will help guide the project's development and ensure that it aligns with real operational conditions.

All information gathered will be kept confidential and used solely for academic purposes.

Thank you very much for your time and continued support.

Sincerely yours,

Lance Christian F. Romero
Project Leader

Public Order and Safety Office
(P.O.S.O.)
Trece Martires City
RECEIVED
BY:
DATE: 10/21/25

Noted:

Alyssa Paola Pocaan, MIT
Thesis 1 Professor

Jerian R. Peren, LPT, PhD
Program Chair, Computer Science

STARS

CAVITE
Governor's Drive, Brgy. Manggahan, City of General Trias,
Cavite, 4107 Philippines
Tel. No.: +6346 481-1400
www.lpu.edu.ph

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APPENDIX N

Consultation Log sheet

Lyceum of the Philippines University
General Trias, Cavite
College of Information Technology and Computer Science
First Semester, A.Y 2025-2026

TECHNICAL ADVISER CONSULTATION LOG SHEET for Thesis 1

Capstone Title: NaviCav: A Web–Mobile Route Planner for Multi-Modal Commuting in Cavite Using Monte Carlo–Guided Genetic Optimization.

Name of Members: Lance Christian F. Romero; Romuel L. Borja; Dyan Paul A. Mercado

Program and Section: BSCS CS401

Prototype	Date and Time of Consultation	No. of Modules Fully Implemented	No. of Modules Partially Implemented	Running Score	Percentage Completed	Research Adviser Signature	Research Teacher Signature
1st Checking						JERIAN R. PEREN, LPT, PhD	ALYSSA PAOLA POCAAN, MIS
Should be within 25% - 50% of the system must be running	**Remarks** System 24.92%						

Prototype	Date and Time of Consultation	No. of Modules Fully Implemented	No. of Modules Partially Implemented	Running Score	Percentage Completed	Research Adviser Signature	Research Teacher Signature
2nd Checking				14/26		JERIAN R. PEREN, LPT, PhD	ALYSSA PAOLA POCAAN, MIS
Should be within 51% - 80% of the system must be running	Remarks System - 53.85% - refer to my post dated 8.28.25 on integrating Carbon Footprint.						

Prototype	Date and Time of Consultation	No. of Modules Fully Implemented	No. of Modules Partially Implemented	Running Score	Percentage Completed	Research Adviser Signature	Research Teacher Signature
3rd Checking				22/26		JERIAN R. PEREN, LPT, PhD	ALYSSA PAOLA POCAAN, MIS
Should be within 100% of the system must be running	Remarks system - 84.62% → for target to 100% prior to defense. - work on remaining modules and on algorithm.						

This is to certify that I have been regularly consulted by my advisees, have reviewed the output system of the above stated study. As their technical adviser, I therefore submit them ready for Proposal Defense as their Prototype is within the required percentage.

Signed:

JERIAN R. PEREN, LPT, PhD
Signature of Research Adviser over printed name

APPENDIX O

Turnitin Result